5.集成运算放大器应用电路

5.1 引言

➢ 集成运算放大器应用极其广泛，总体分为两大类：

- 线性应用
- 非线性应用

➢ 把运放看作理想运放，使分析过程简化。

理想运放：

将集成运放的各项技术指标理想化后的一种最简模型

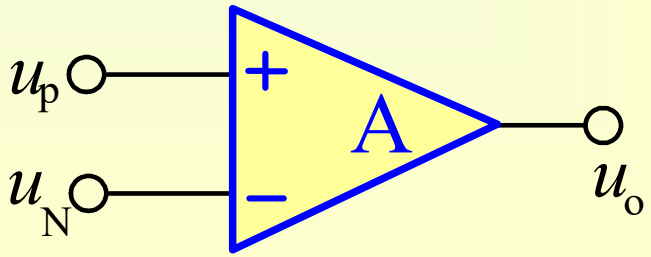
理想运放的主要技术指标

(1) $A_{od}=\infty$

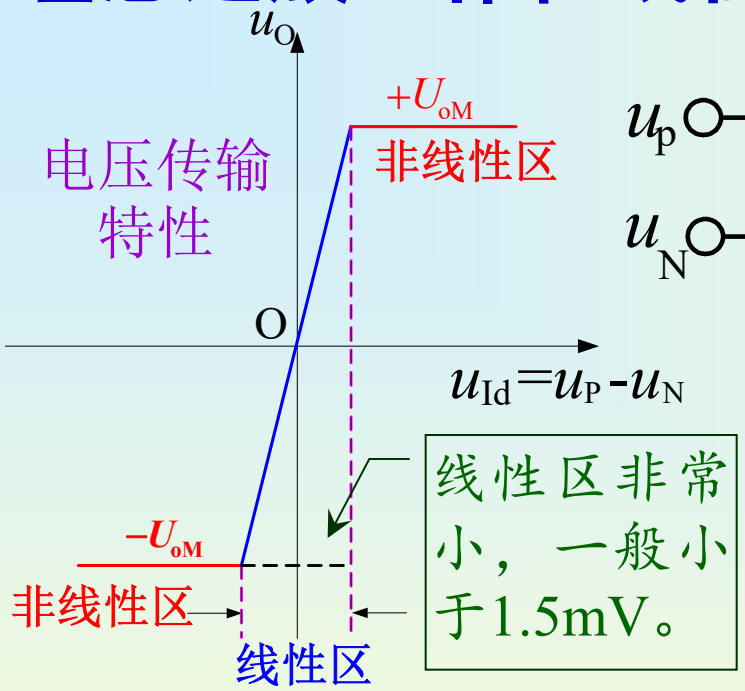
(2) $R_{id}=\infty, R_o=0$

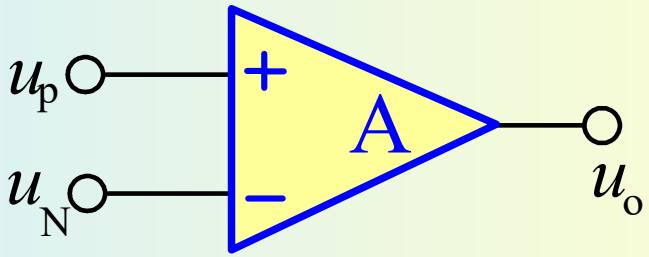
(3) $K_{CMR}=\infty$

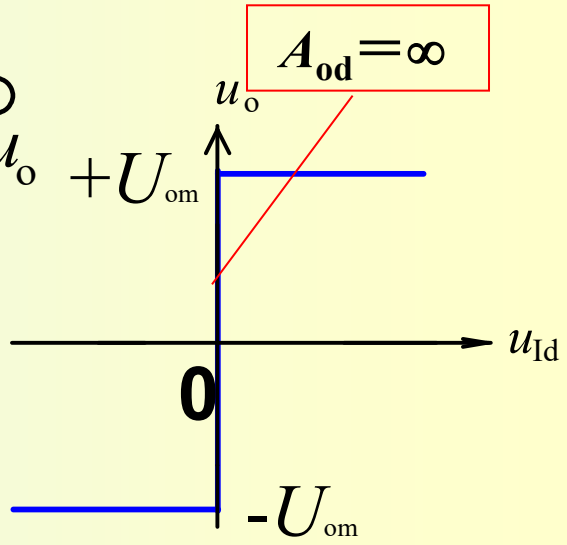
(4) $BW=\infty$



理想运放工作在线性区和非线性区的特点







线性区： $u_{Id}\approx 0\text{V}$　$u_P\approx u_N$　$i_P\approx i_N\approx 0$　虚短、虚断

运算、滤波

非线性区：

电压比较器

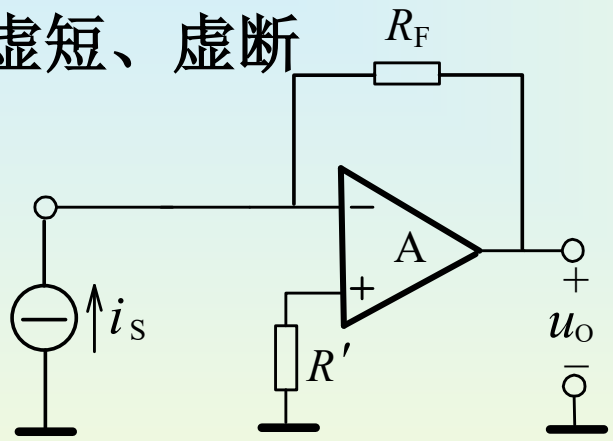
$$u_O=\begin{cases}+U_{om} & (u_P>u_N)\\ -U_{om} & (u_P<u_N)\end{cases}$$

$i_P\approx i_N\approx 0$　只有虚断

用“虚短虚断”可以求解线性区的运算、变换关系式

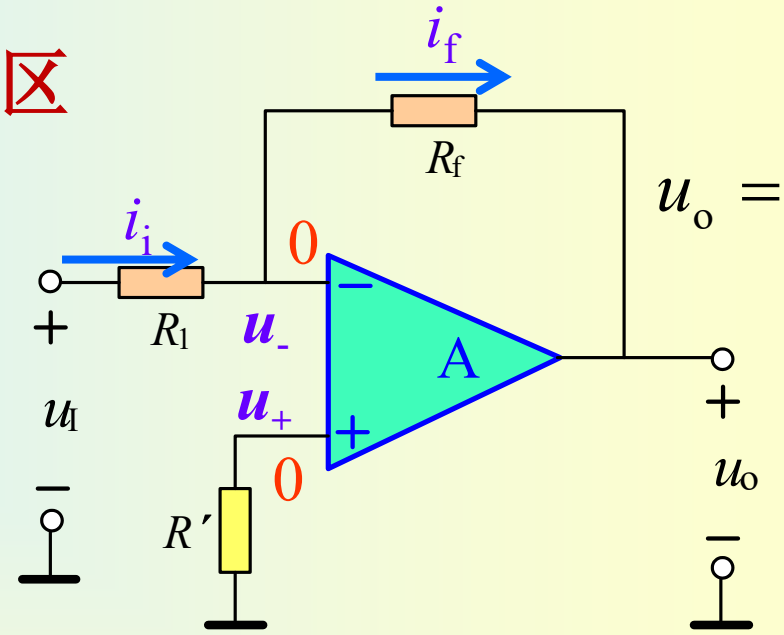
➢ 集成运放工作在线性区

- 深度负反馈
- 虚短、虚断



电流→电压变换器(互阻放大器)

$$u_O = A_R i_S = -i_S R_F$$



$$u_o = -\frac{R_f}{R_1} u_i$$

反相比例运算

➢ 时域：

- 运算：比例、加减、乘除、微分、积分、对数、指数…
- 变换：电压-电流、电流-电压、阻抗变换…

➢ 频域：滤波

运算在上学期已学，本节只讲滤波电路

5.2有源滤波器

滤波器的功能：对频率进行选择，过滤掉噪声和干扰信号，保留下有用信号。

有源滤波器实际上是一种具有特定频率响应的放大器。

滤波器的分类：

低通滤波器（LPF）

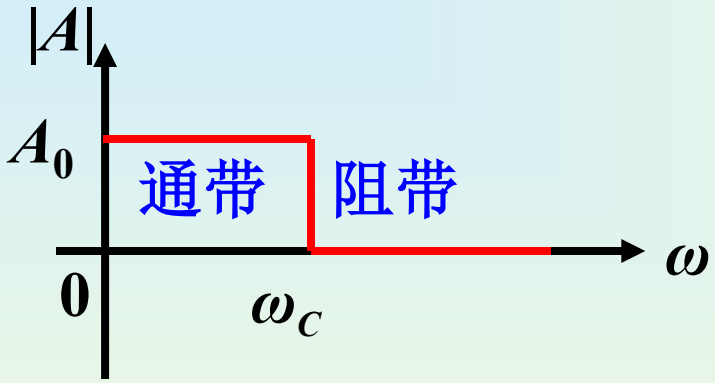
高通滤波器（HPF）

带通滤波器（BPF）

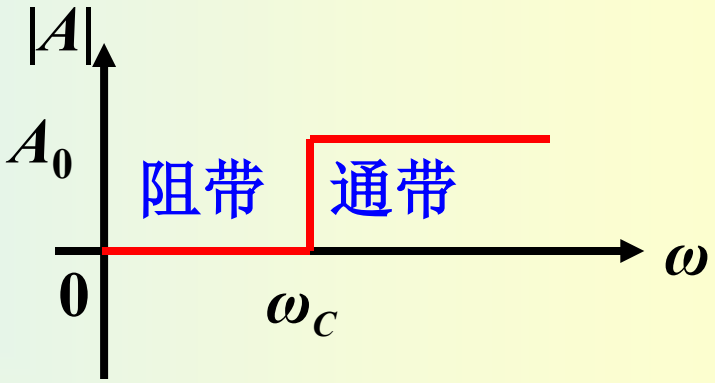
带阻滤波器（BEF）

各种滤波器理想的幅频特性

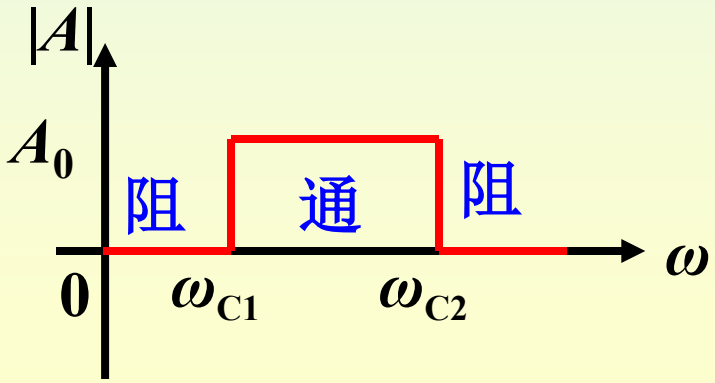
（1）低通



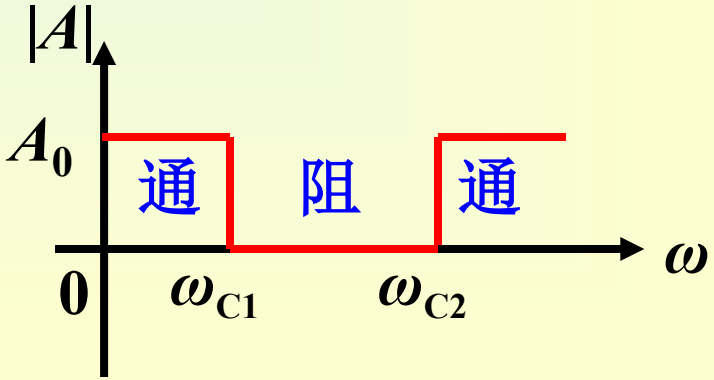
（2）高通



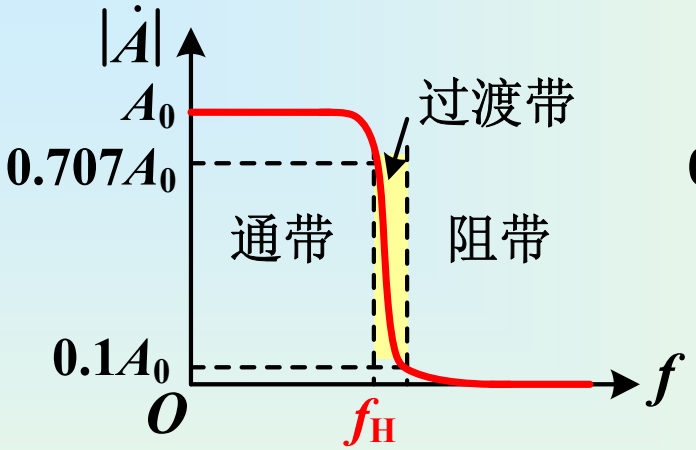
（3）带通

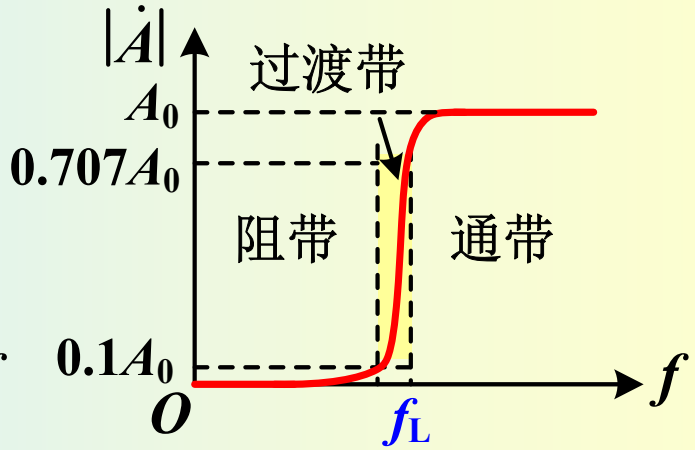


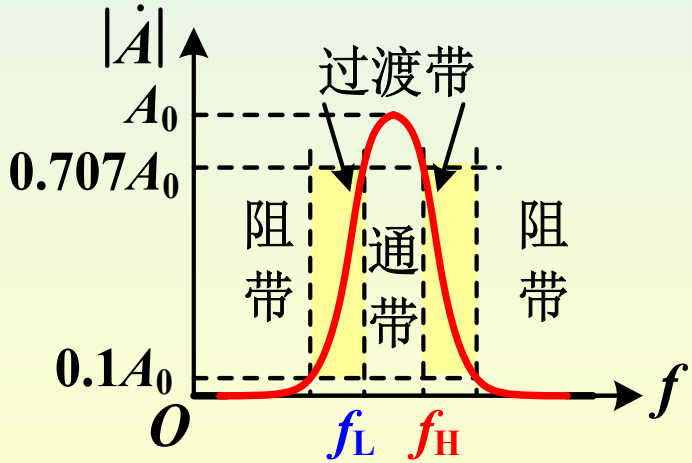
（4）带阻

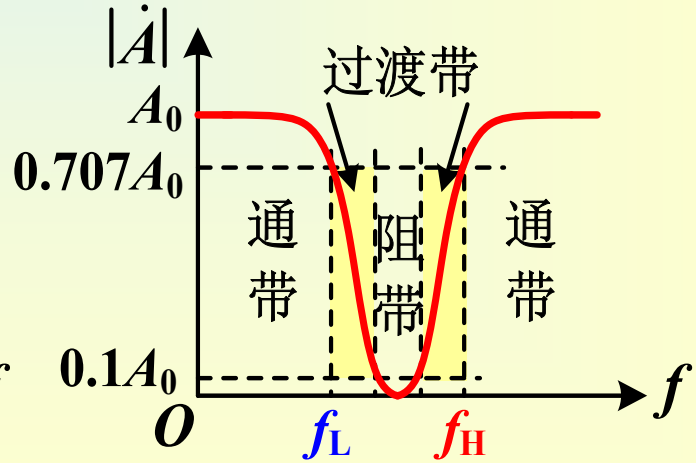


实际滤波器的幅度频率特性

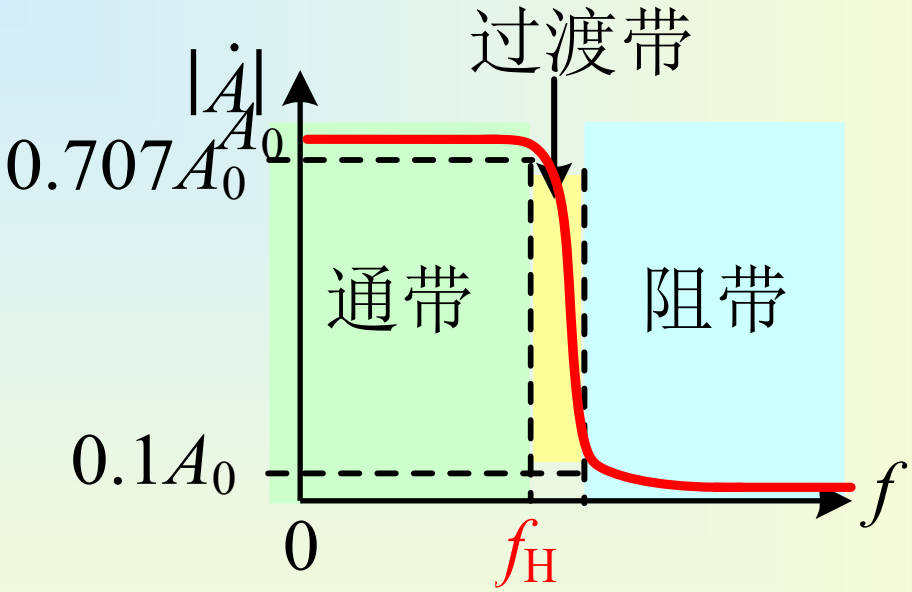








1. 低通有源滤波器



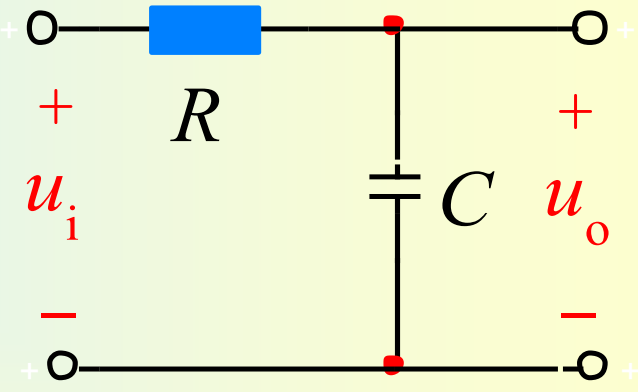
低通滤波器的幅频特性

过渡带频段内的信号虽然被抑制，但是仍然能通过滤波器；因而在设计滤波器时，要求通带之外的增益下降速度越快越好，即滤波器的过渡带越窄，电路的滤波特性越理想。

(1) 一阶RC低通滤波器(无源)

传递函数:

$$A_u(s)=\frac{u_o(s)}{u_i(s)}=\frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}}=\frac{1}{1+j\omega RC}=\frac{1}{1+j\frac{\omega}{\omega_0}}$$



特征频率: $\omega_0=\frac{1}{RC}$

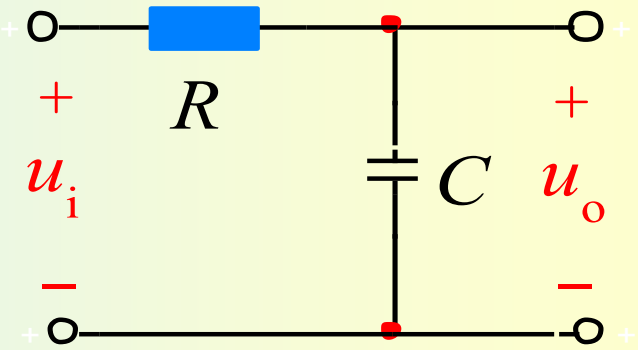
把$j\omega$换为S，则拉普拉斯(复频域)表示为:

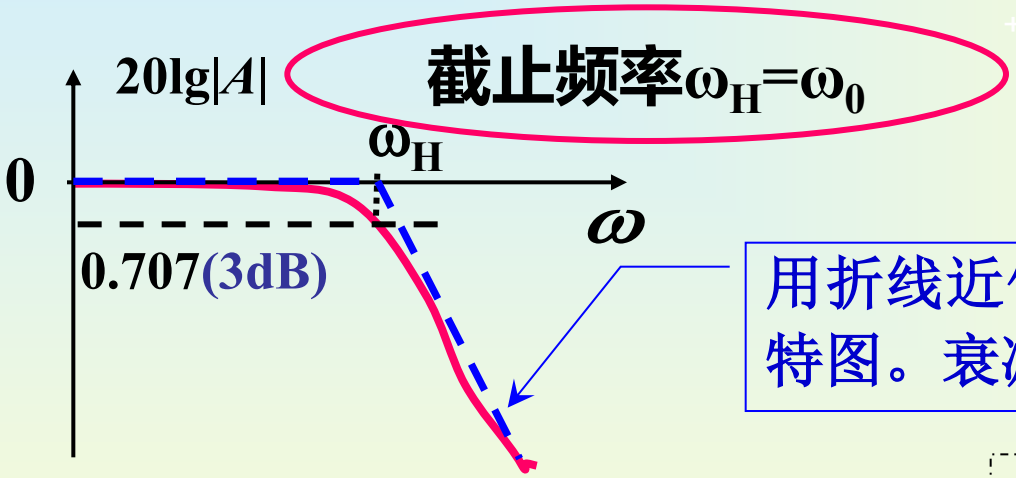
幅频特性: $|A(\omega)|=\frac{1}{\sqrt{1+(\frac{\omega}{\omega_0})^2}}$

$$\frac{1/RC}{S+1/RC}=\frac{a}{S+a}$$

一阶滤波

幅频特性： $|A|=\dfrac{1}{\sqrt{1+(\dfrac{\omega}{\omega_0})^2}}$





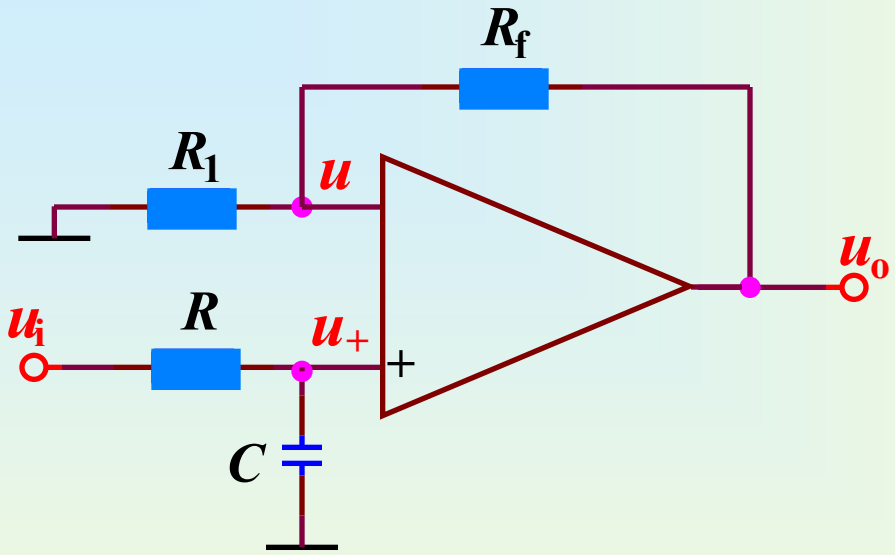
用折线近似描述幅频变化趋势称为波特图。衰减速率-20dB/十倍频程。

此电路的缺点：

1、带负载能力差。
2、无放大作用。
3、特性不理想，边沿不陡。

当$\omega>>\omega_0$时，
$20\log|A|\approx-20\log(\omega/\omega_0)$，
每增加十倍频下降-20dB。
表示-20dB/十倍频程，或者表示-20dB/dec。

（2）一阶有源低通滤波器



$$u_+ = \left(\frac{1}{1+j\omega RC}\right)u_i$$

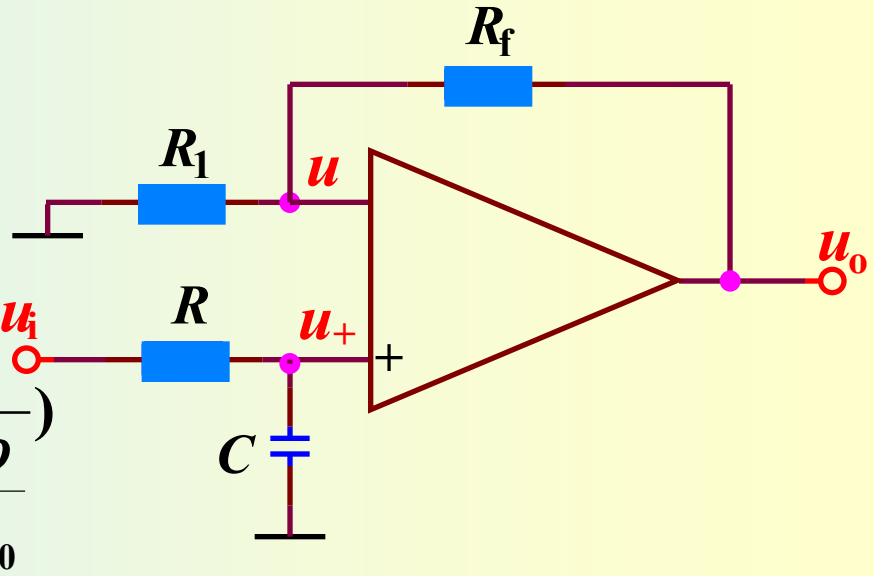
$$u_o = \left(1+\frac{R_f}{R_1}\right)u_+$$

$$u_o = \left(1+\frac{R_f}{R_1}\right)\left(\frac{1}{1+j\omega RC}\right)u_i$$

传递函数： $A(\omega)=\dfrac{u_o(\omega)}{u_i(\omega)}=\left(1+\dfrac{R_f}{R_1}\right)\left(\dfrac{1}{1+j\omega RC}\right)=\left(\dfrac{A_0}{1+j\dfrac{\omega}{\omega_0}}\right)$

通带增益： $A_0=1+\dfrac{R_f}{R_1}$　　**特征频率：** $\omega_0=\dfrac{1}{RC}$

幅频特性及幅频特性曲线



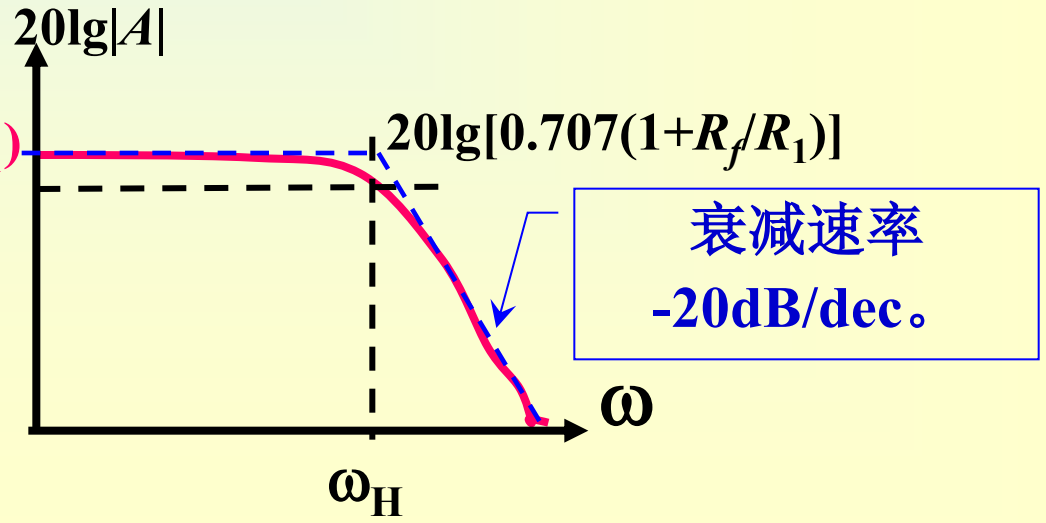
传递函数：

$$A(\omega)=\frac{u_o(\omega)}{u_i(\omega)}=(1+\frac{R_f}{R_1})(\frac{1}{1+j\frac{\omega}{\omega_0}})$$

幅频特性：

$$|A(\omega)|=(1+\frac{R_f}{R_1})\frac{1}{\sqrt{1+(\frac{\omega}{\omega_0})^2}}$$

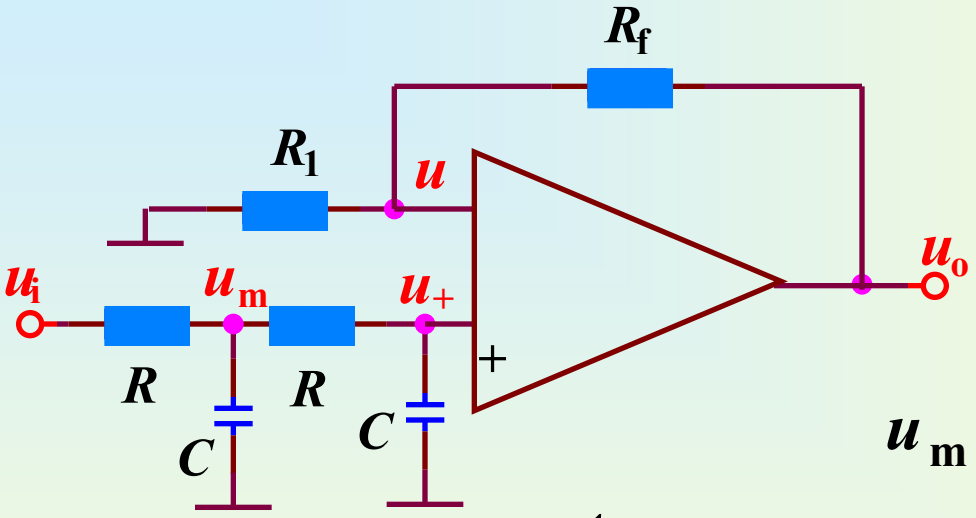
截止频率：$\omega_H=\omega_0=\frac{1}{RC}$



缺点：阻带衰减太慢。

(3) 二阶有源低通滤波器

①简单二阶低通滤波电路



$$u_o = \left(1+\frac{R_f}{R_1}\right)u_+$$

$$u_+ = u_m \frac{1}{1+j\omega CR}$$

$$u_m = \frac{\frac{1}{j\omega C}//\left(R+\frac{1}{j\omega C}\right)}{R+\left[\frac{1}{j\omega C}//\left(R+\frac{1}{j\omega C}\right)\right]}u_i$$

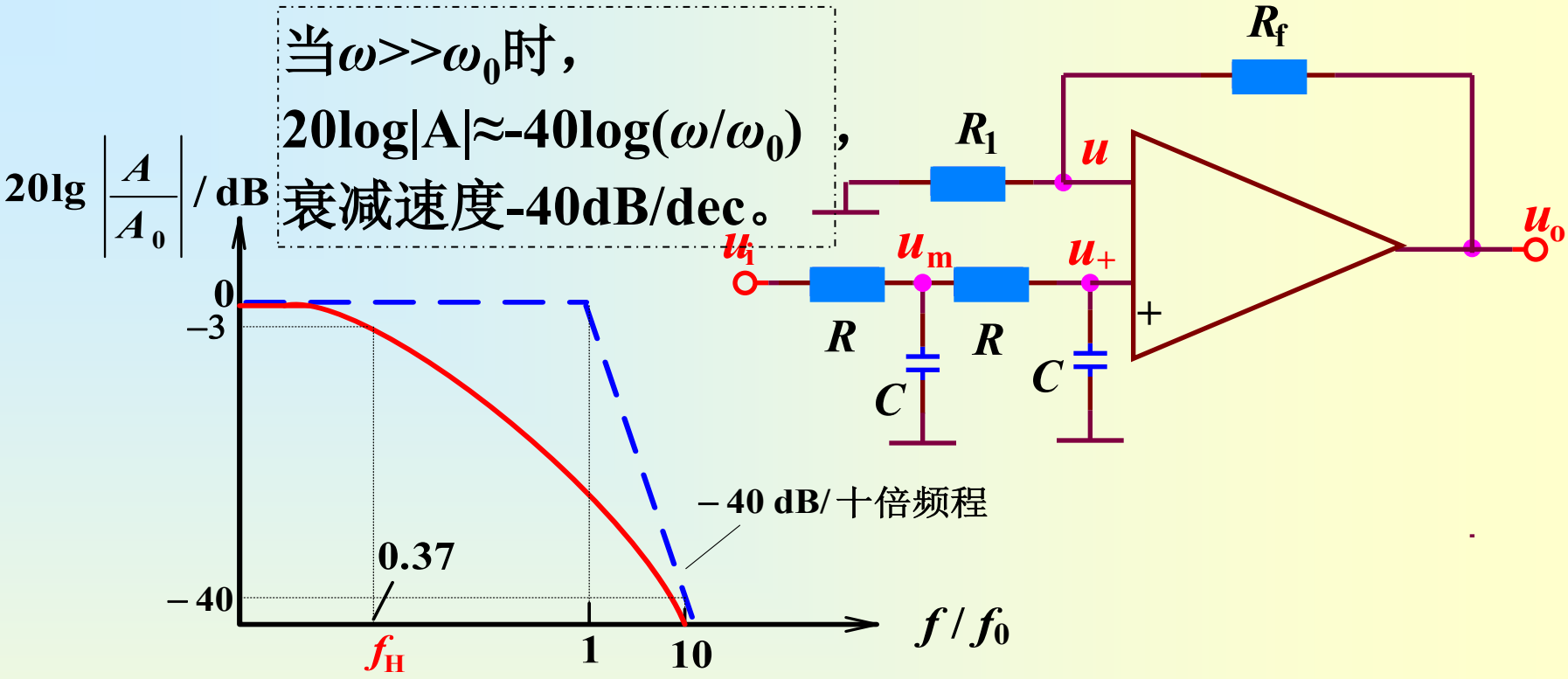
$$A=\frac{u_o}{u_i}=\frac{A_0}{1+3\,j\omega CR+(j\omega CR)^2}$$

通带增益： $A_o = 1+\dfrac{R_f}{R_1}$

把$j\omega$换为S，则拉普拉斯(复频域)可表示为：

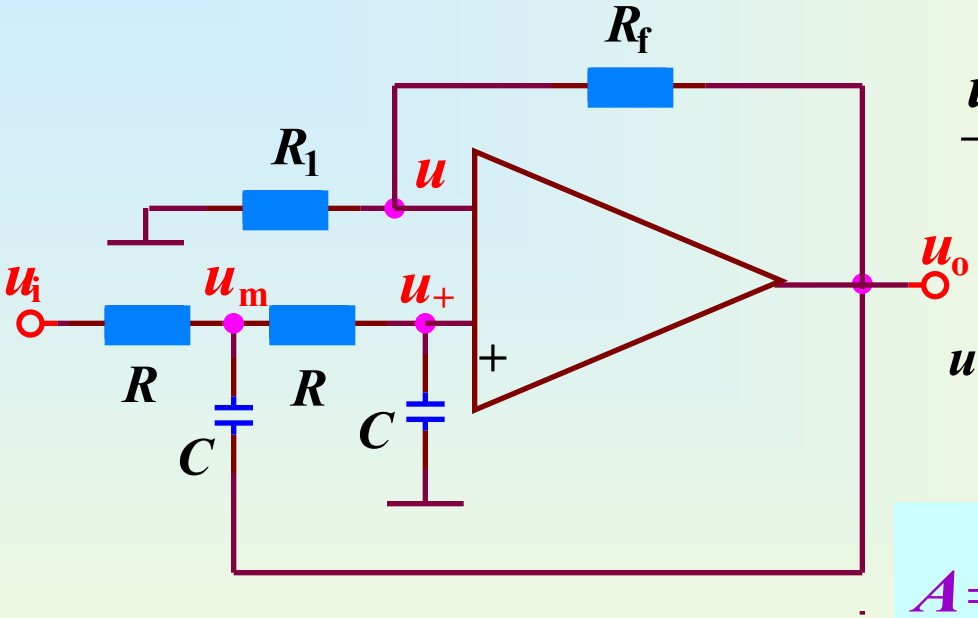
$$A=\frac{a_0}{b_0+b_1S+b_2S^2}=\frac{a_0}{(S+\alpha)(S+\beta)}$$

有二个极点，即二阶滤波。



与理想的二阶波特图相比，在超过f_0以后，幅频特性以-40dB/dec的速率下降，比一阶的下降快。但在通带截止频率$f_H \rightarrow f_0$之间幅频特性下降的还不够快，幅频特性下降缓慢，过渡带太长。如果在$f \rightarrow f_0$区间适当提升电压放大倍数，即可拉近f_H与f_0，减小过渡带且通带不至于过度减小。

②二阶压控型低通滤波电路



$$u_o=\left(1+\frac{R_f}{R_1}\right)u_+=A_0\cdot u_+$$

$$\frac{u_i-u_m}{R}=\frac{u_m-u_+}{R}+\frac{u_m-u_o}{\frac{1}{j\omega C}}$$

$$u_+=\frac{\frac{1}{j\omega C}}{R+\frac{1}{j\omega C}}\cdot u_m$$

$$A=\frac{a_0}{b_0+b_1(j\omega)+b_2(j\omega)^2}$$

传递函数：

$$A=\frac{u_o}{u_i}=\frac{A_0}{1+(3-A_0)j\omega CR+(j\omega CR)^2}$$

$$=\frac{A_0}{1-\left(\frac{\omega}{\omega_0}\right)^2+j\left(\frac{\omega}{\omega_0 Q}\right)}$$

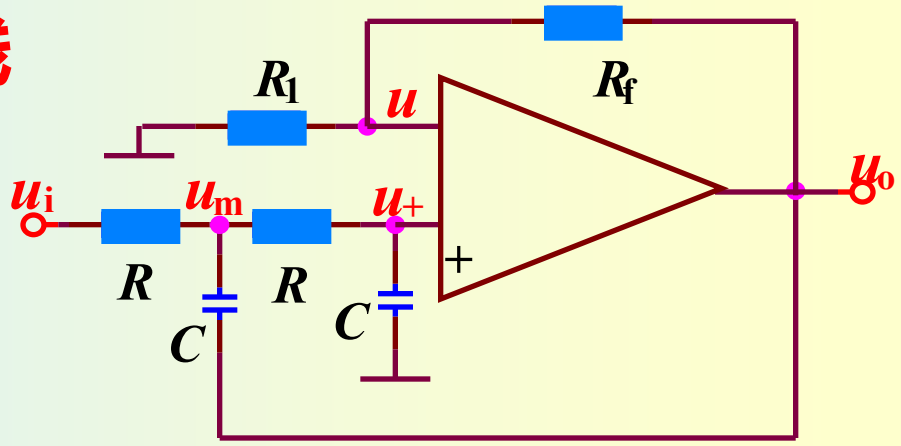
$$Q=\frac{1}{3-A_0}$$

$$\omega_0=\frac{1}{RC}$$

幅频特性及幅频特性曲线

$$|A|=\frac{A_0}{\sqrt{[1-(\frac{\omega}{\omega_0})^2]^2+(\frac{\omega}{\omega_0 Q})^2}}$$

$$Q=\frac{1}{3-A_0}$$



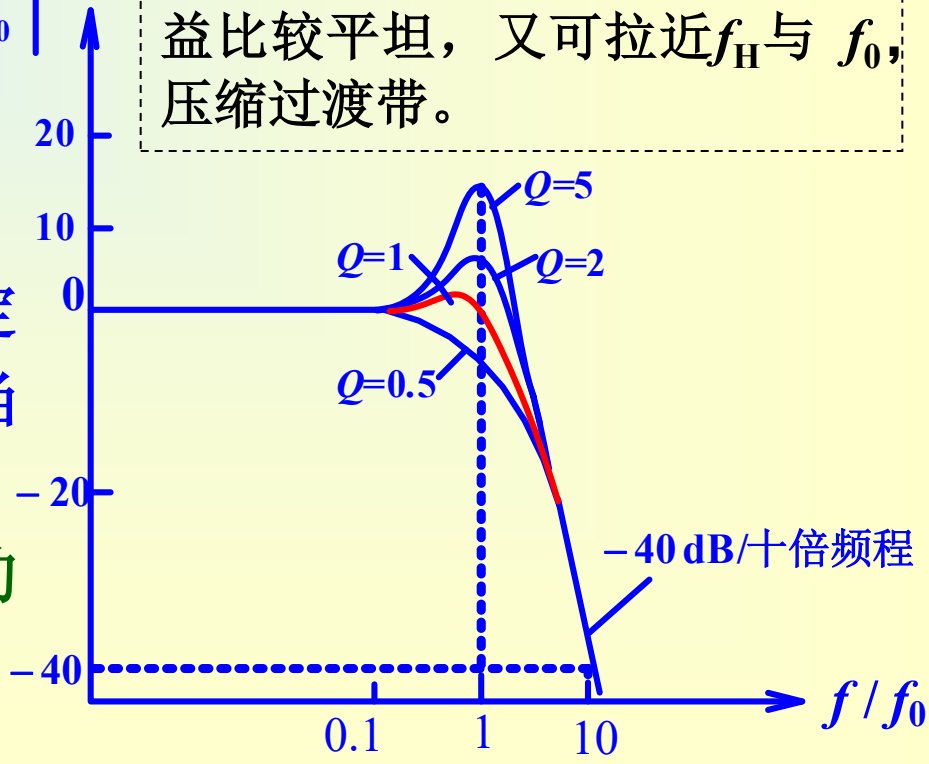
$20\lg\left|\frac{A}{A_0}\right|$/dB

一般选择$Q\leq1$，其通带内的增益比较平坦，又可拉近f_H与 f_0，压缩过渡带。

当$A_0\geq3$时，滤波器自激！

当$A_0<3$时，滤波器可以稳定工作。此时特性与Q有关。当$Q=0.707$时，幅频特性平坦。

当$f>>f_0$时，幅频特性曲线的斜率为-40dB/dec。



2. 高通有源滤波器

（1）低通滤波器和高通滤波器对偶关系

低通

R

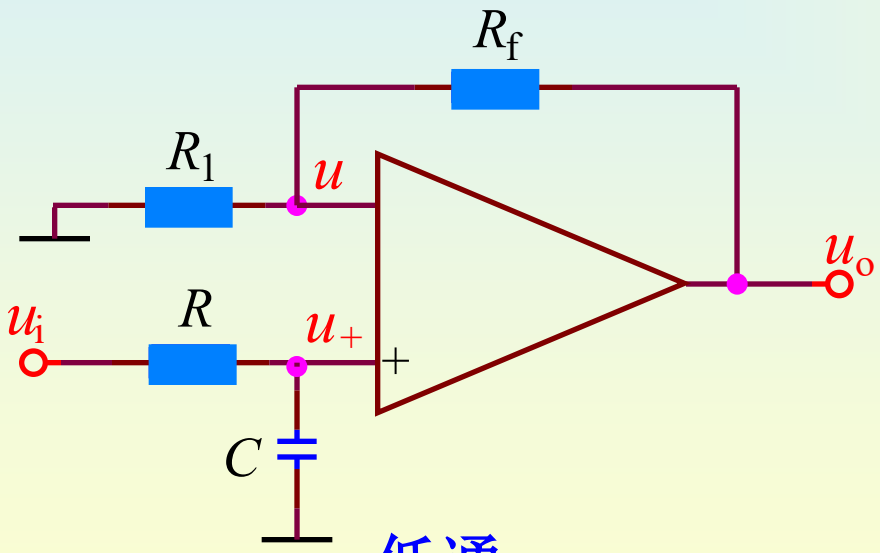
C

滤波电路对偶

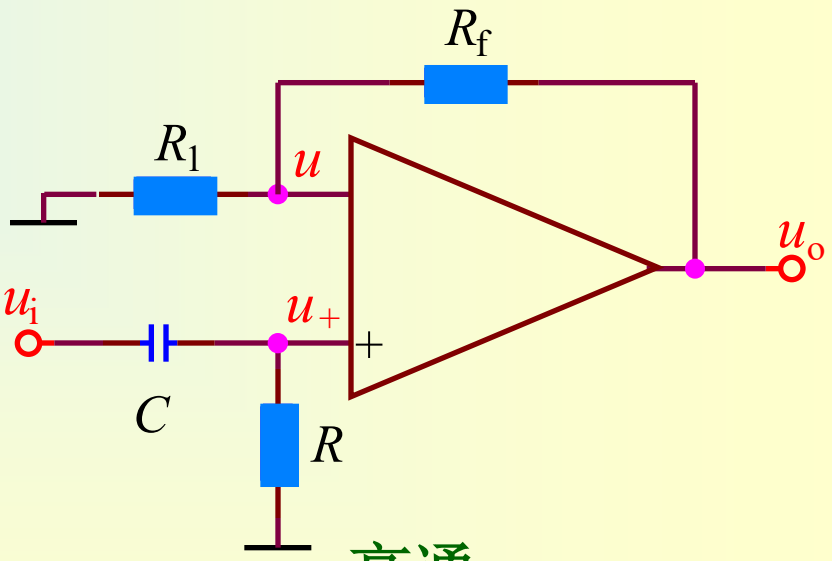
高通

C

R



低通



高通

R_x

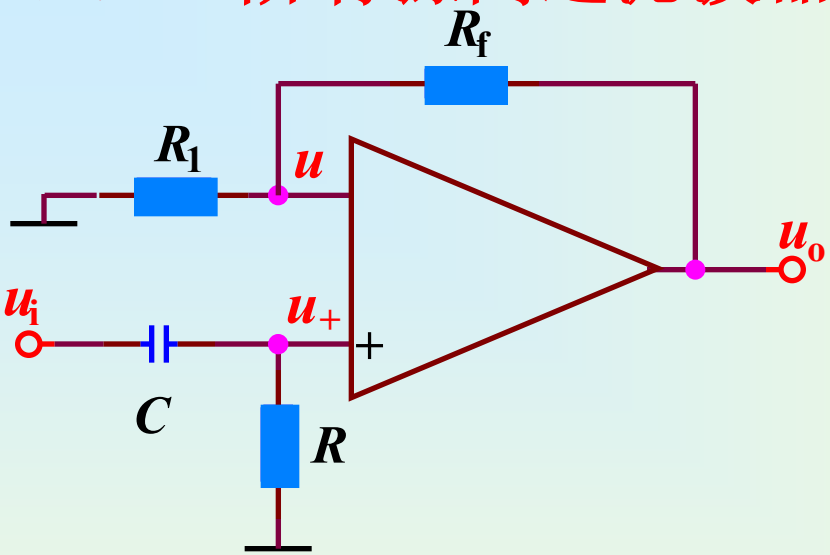
$(1/j\omega C_y)$

传递函数对偶

$(1/j\omega C_x)$

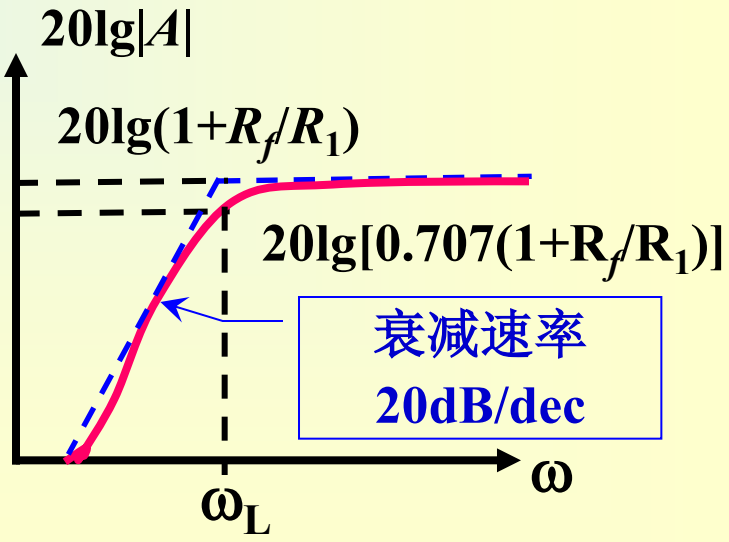
R_y

（2）一阶有源高通滤波器



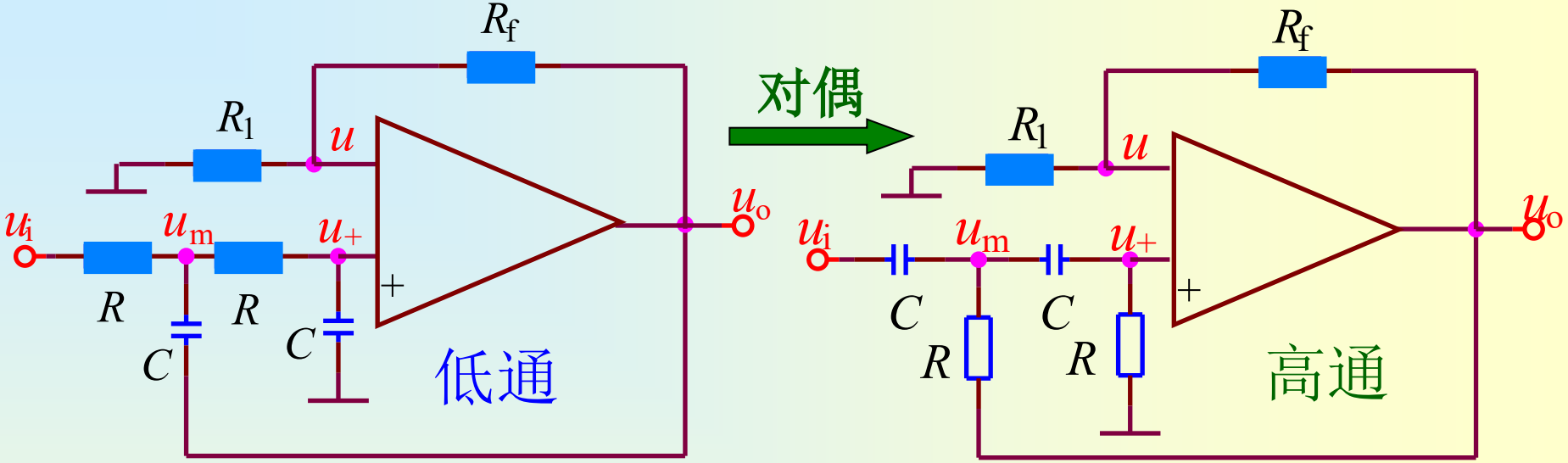
通带增益：$A_o = 1 + \dfrac{R_f}{R_1}$

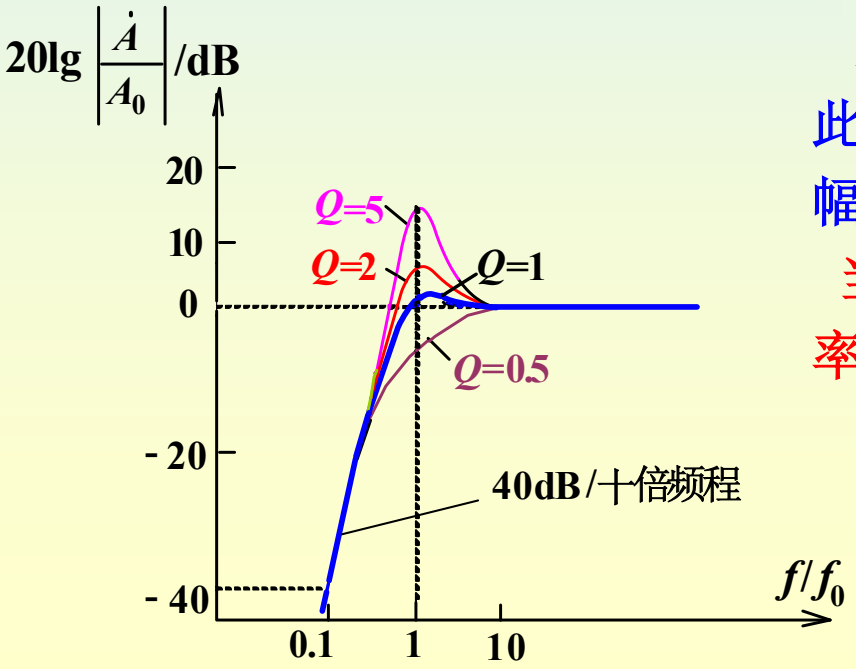
特征频率：$\omega_0 = \dfrac{1}{RC}$



缺点：阻带衰减太慢。

（3）二阶有源高通滤波器





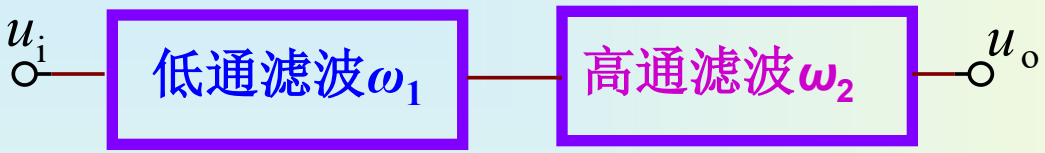
当A_0＜3时，滤波器可以稳定工作。此时特性与Q有关。当Q=0.707时，幅频特性较平坦。

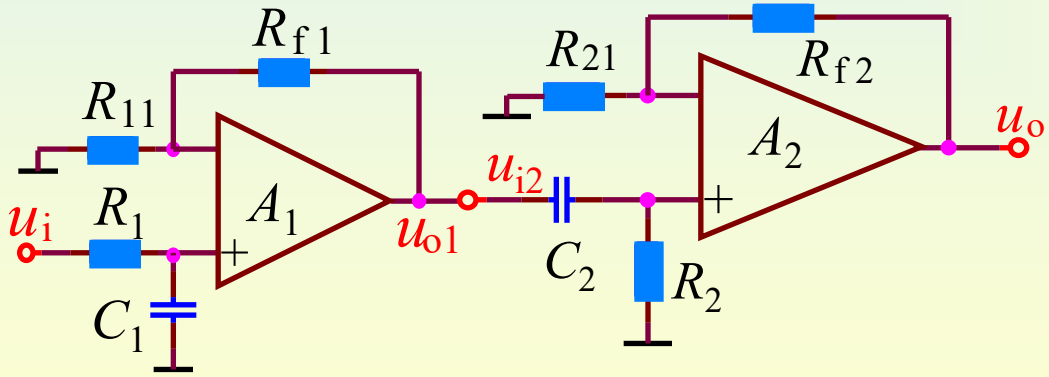
当f ＜＜ f_0时，幅频特性曲线的斜率为+40dB/dec。

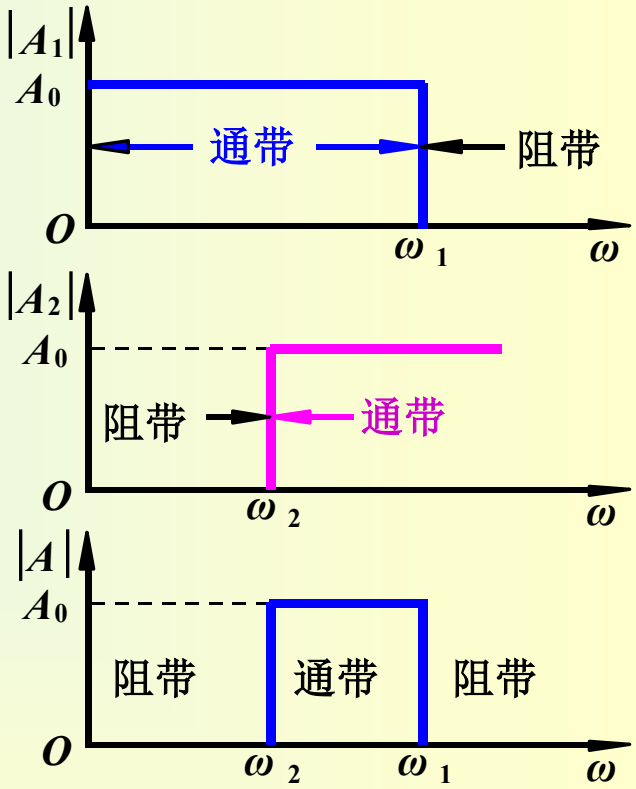
当A_0≥3时，电路自激。

3. 有源带通滤波器

可由低通和高通串联得到





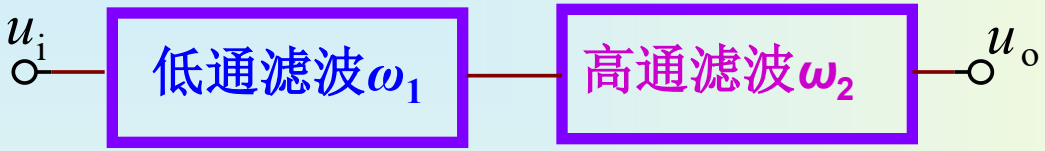


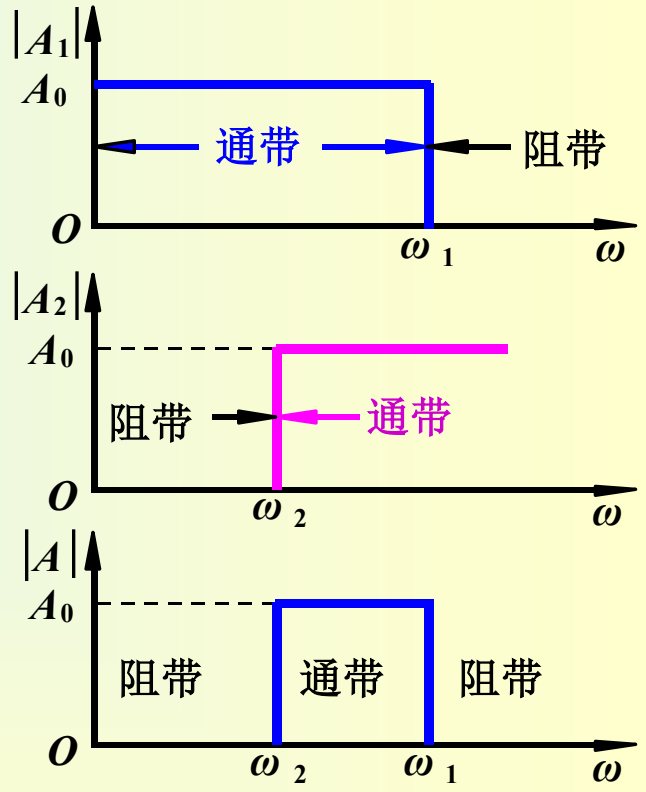
（宽带）带通滤波器

必须满足 $\omega_2 < \omega_1$

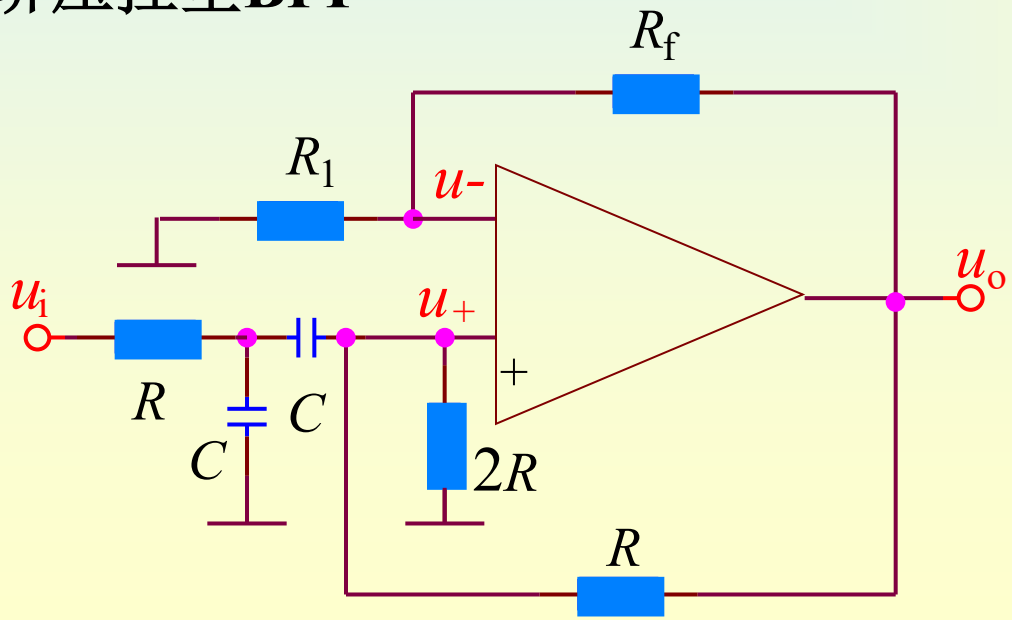
3. 有源带通滤波器

可由低通和高通串联得到



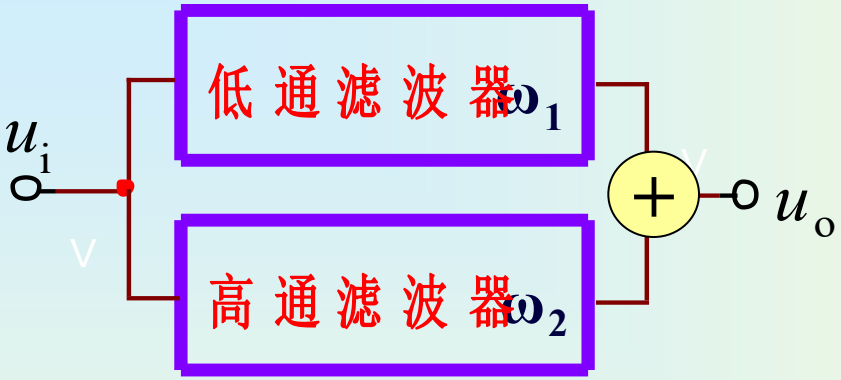


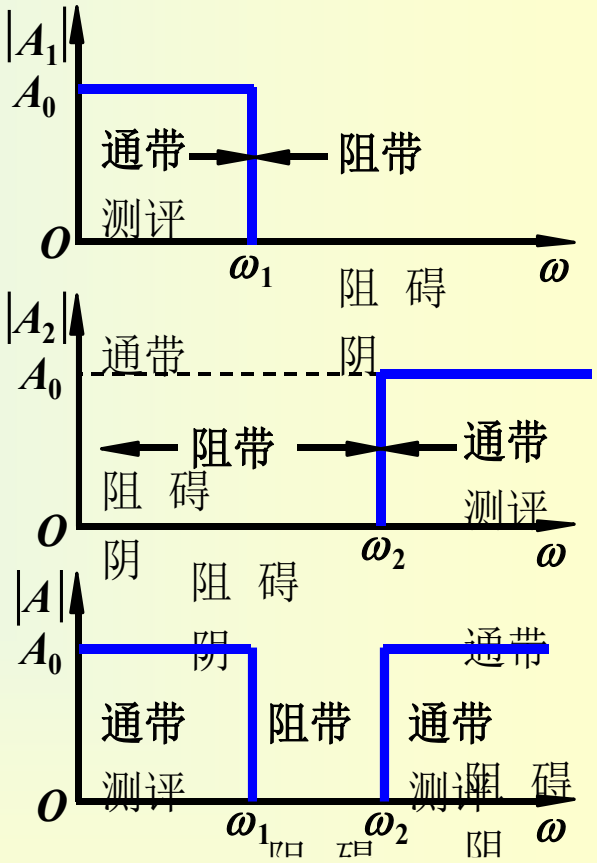
为获得较窄的通频带：
二阶压控型BPF



必须满足 $\omega_2 < \omega_1$

4. 有源带阻滤波器



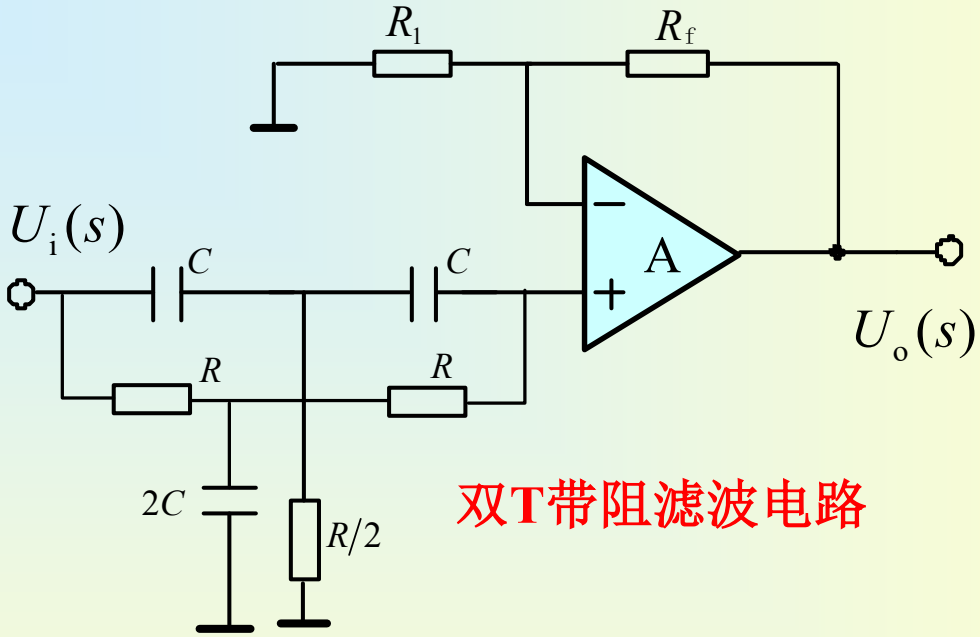


由低通和高通并联求和可实现带阻滤波器

必须满足 $\omega_2 > \omega_1$

4. 有源带阻滤波器

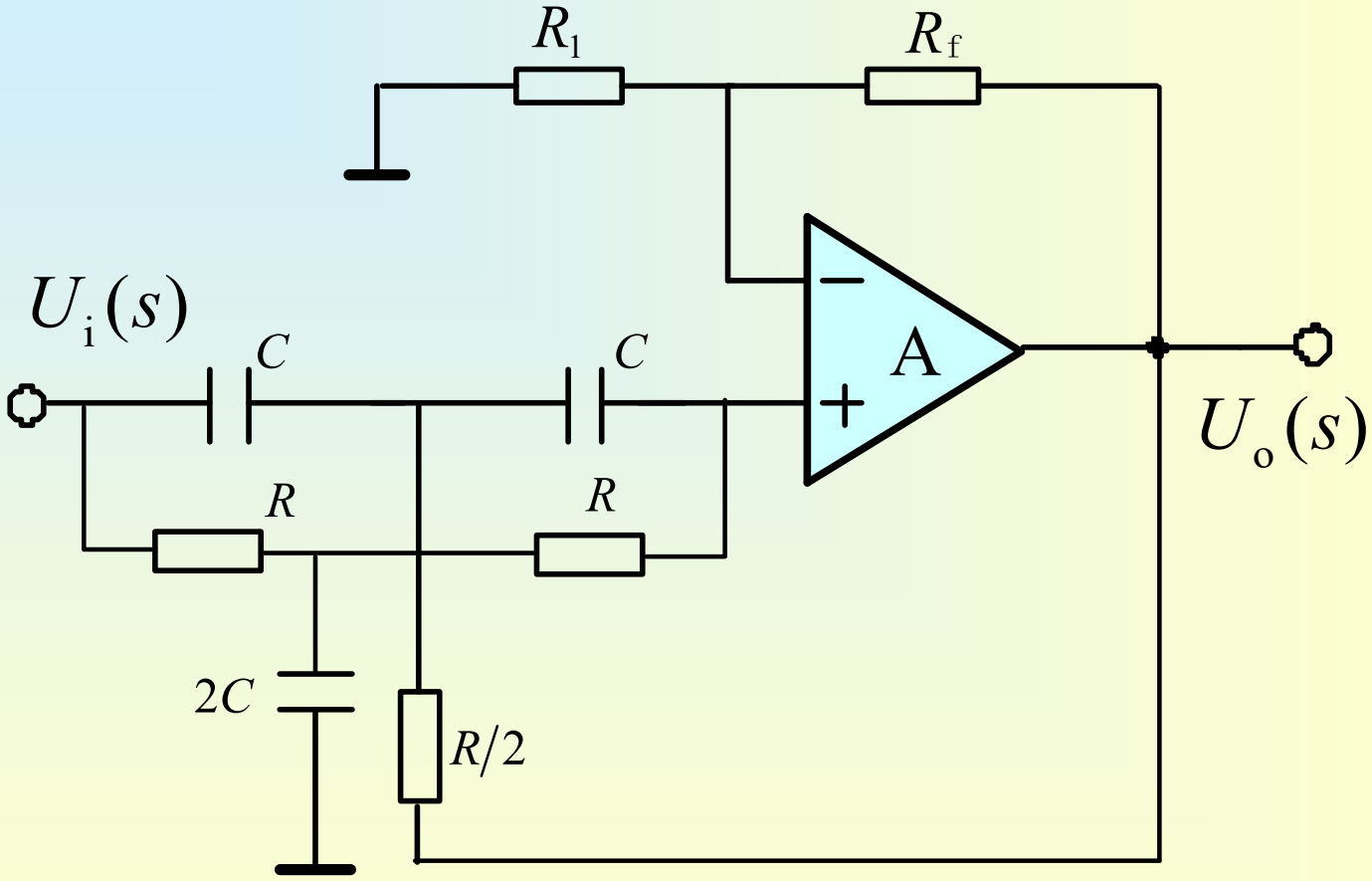
无源带阻滤波电路+同相比例运算电路=有源带阻滤波电路



双T带阻滤波电路

当输入高频信号时，电容的容抗值小，可近似视作“短路”，高频信号由上方的两个电容和一个电阻所构成的T 型网络通过；当输入低频信号时，电容的容抗值大，可近似视作“开路”，低频信号由下方的两个电阻和一个电容所构成的 T 型网络通过；而处于低频和高频之间的频带信号无法通过双T 网络。

4. 有源带阻滤波器



二阶压控型有源带阻滤波器

有源滤波器小结

- 有源滤波器是一种重要的信号处理电路，它可以突出有用频段的信号，衰减无用频段的信号，抑制干扰和噪声信号，达到选频和提高信噪比的目的。实际使用时，应根据具体情况选择低通、高通、带通或带阻滤波器，并确定滤波器的具体形式。

- 运算电路与滤波电路的异同

相同：均引入深度负反馈，运放工作在线性区，均具有“虚短”和“虚断”的特点，可用节点电流法求解电路。

不同：①运算研究的是时域问题，滤波研究的是频域问题；测试时，前者是在输入信号频率不变或直流信号下测量输出电压与输入电压的关系，后者是在输入电压幅值不变的情况下测量。

②运算电路用运算关系式描述输出电压与输入电压的关系，有源滤波器用传递函数的幅频特性描述滤波特性。

有源滤波器 内容要求

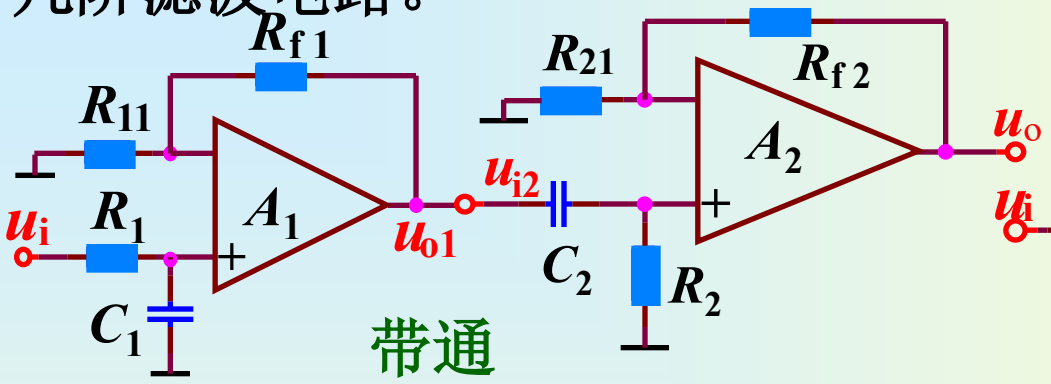
- 识别滤波电路即可

（内容以课件为主，识别课件中讲过的滤波器即可，习题可不做）

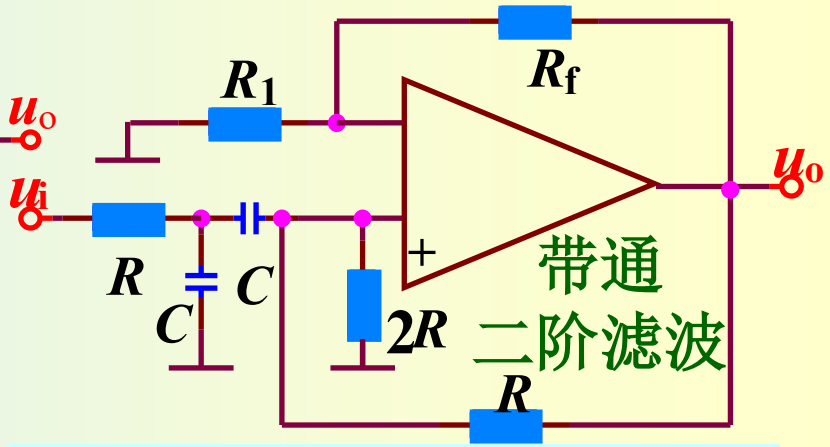
自我检测：二版三版相同，第9章1-3

习题： 二版三版相同，第9章1-5

练习：定性分析下列各电路属于哪种类型滤波电路，并说明是几阶滤波电路。

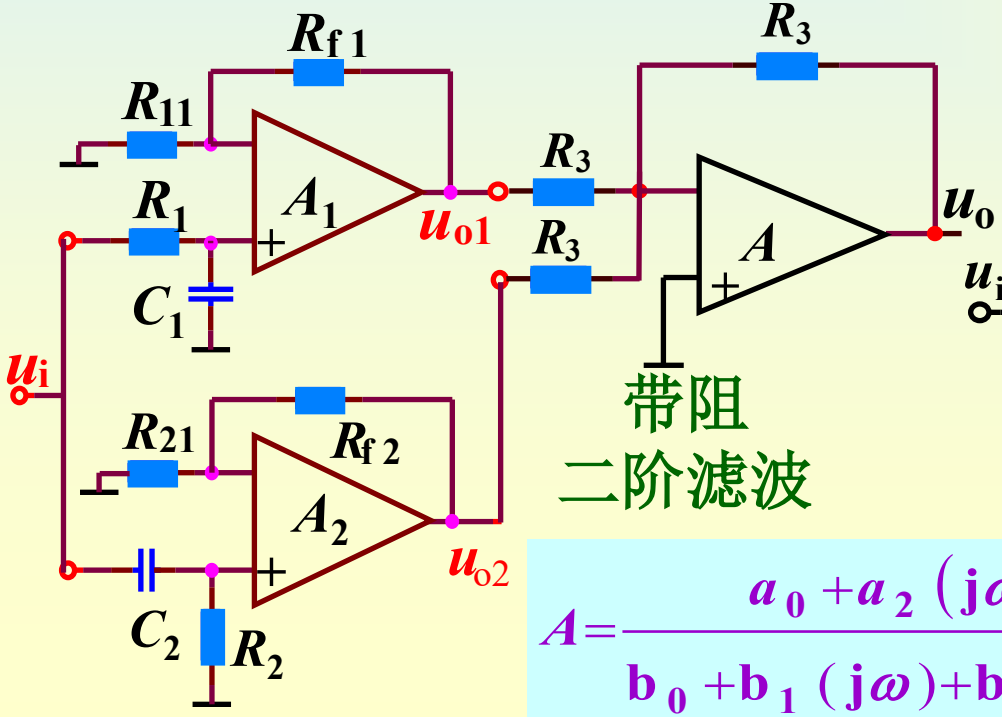


带通
二阶滤波

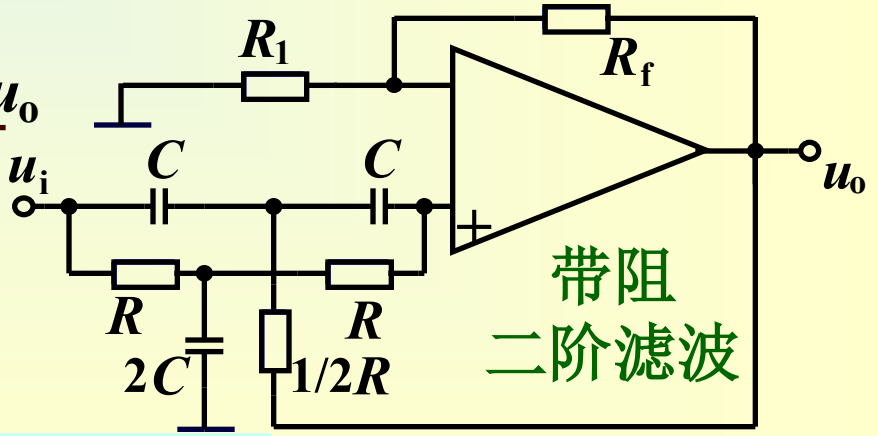


带通
二阶滤波

$$A=\frac{a_1(j\omega)}{b_0+b_1(j\omega)+b_2(j\omega)^2}$$



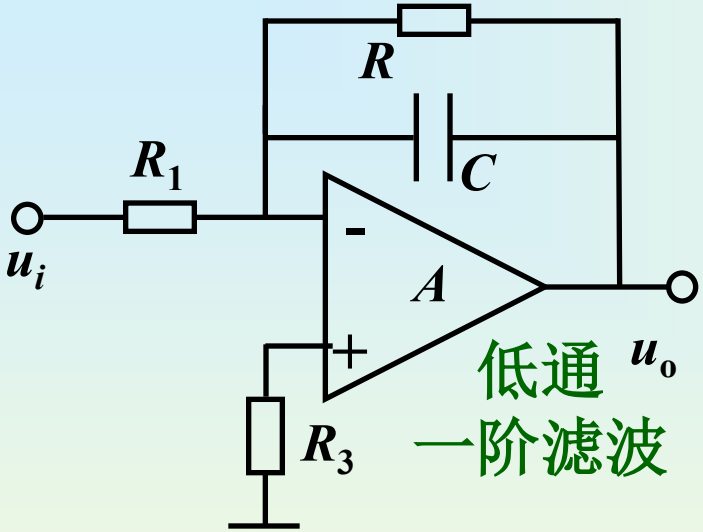
带阻
二阶滤波



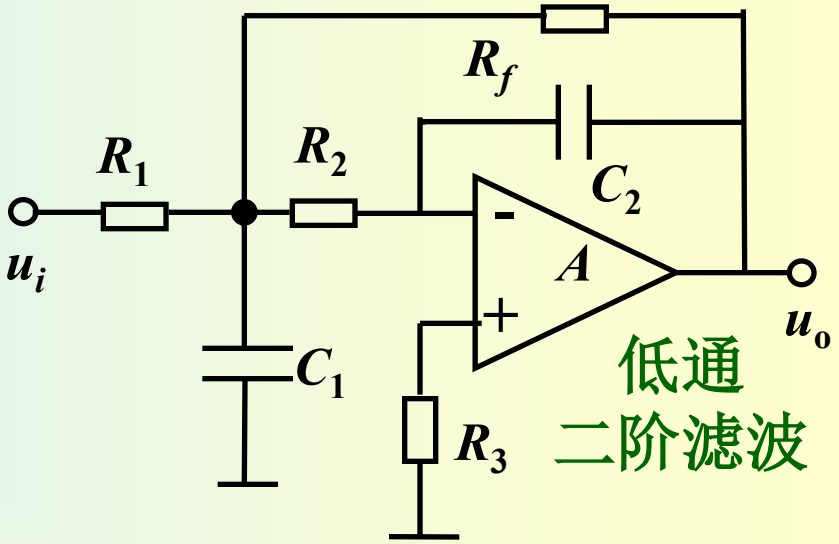
带阻
二阶滤波

$$A=\frac{a_0+a_2(j\omega)^2}{b_0+b_1(j\omega)+b_2(j\omega)^2}$$

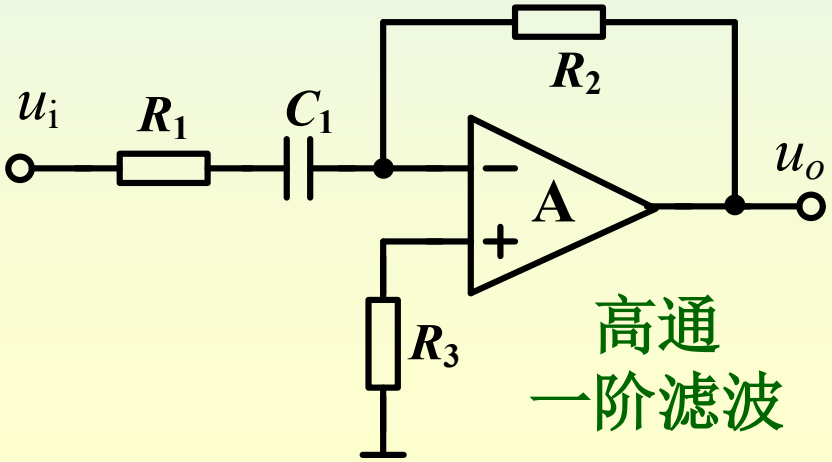
练习（续）：定性分析下列各电路属于哪种类型滤波电路，并说明是几阶滤波。



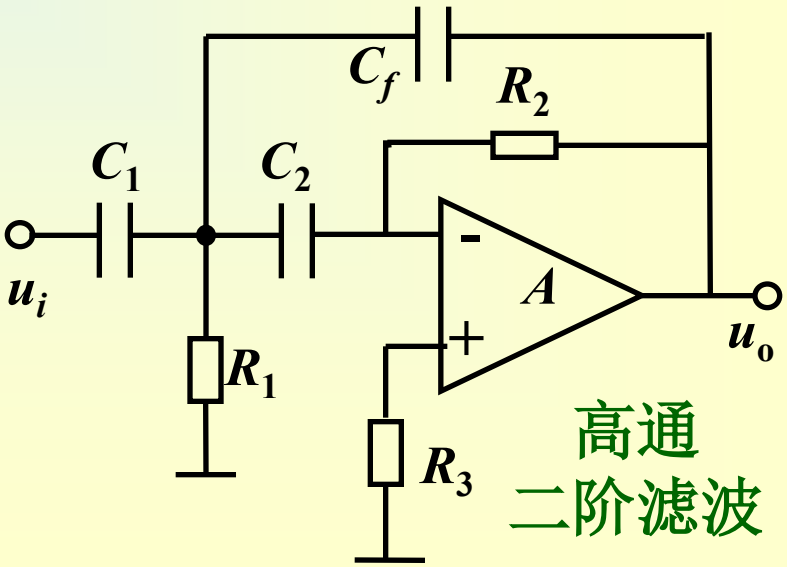
低通
一阶滤波



低通
二阶滤波



高通
一阶滤波



高通
二阶滤波

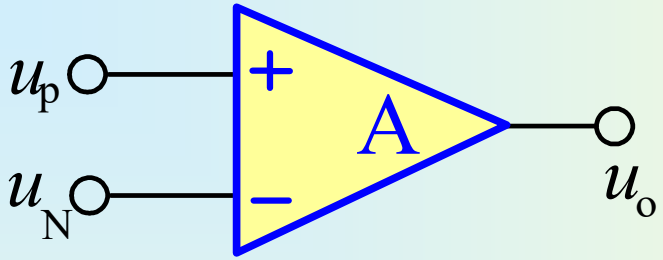
5.3 集成运算放大器非线性应用

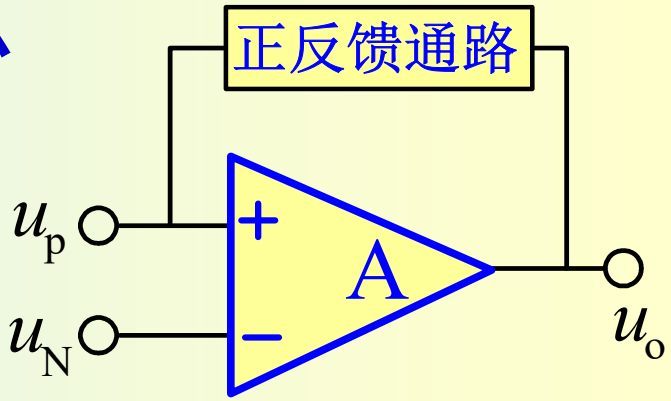
★单限比较器

★滞回比较器

★非正弦波发生器（矩形波、三角波、锯齿波）

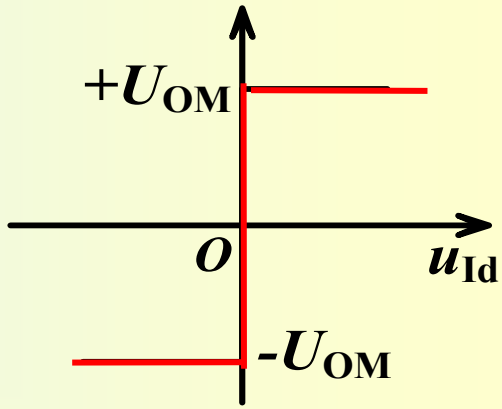
5.3.1 电压比较器电路特点





集成运放开环或正反馈时

$$u_o=\begin{cases}+U_{OM}, & \text{当} u_P > u_N \\ -U_{OM}, & \text{当} u_P < u_N \\ \text{转折区}, & \text{当} u_P \approx u_N\end{cases}$$

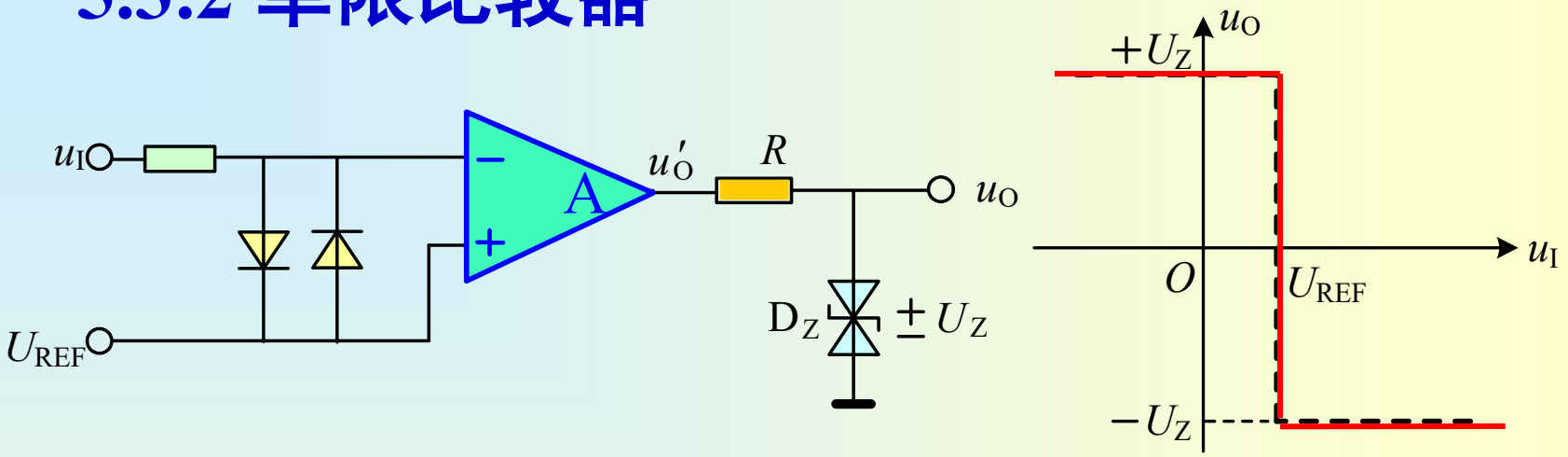


开环——单限比较器

开环——窗口比较器（不在大纲内，课上不讲）

正反馈——滞回比较器

5.3.2 单限比较器



单限比较器只有一个阈值电压

1) 当$u_I < U_{REF}$时，$u_O = +U_Z$

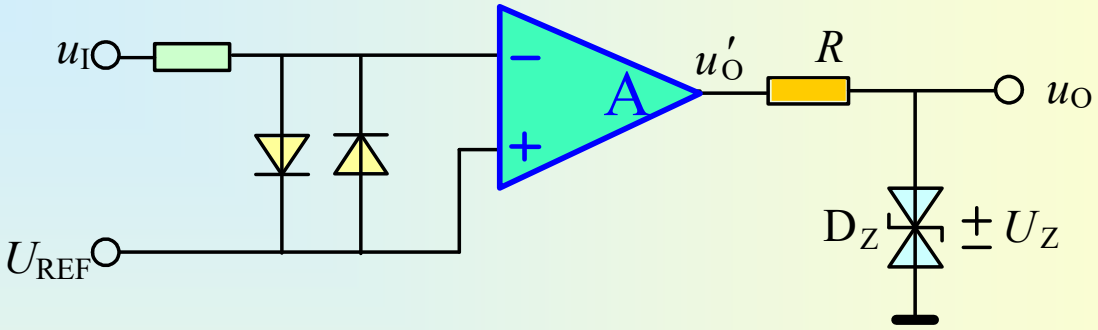
当$u_I > U_{REF}$时，$u_O = -U_Z$;

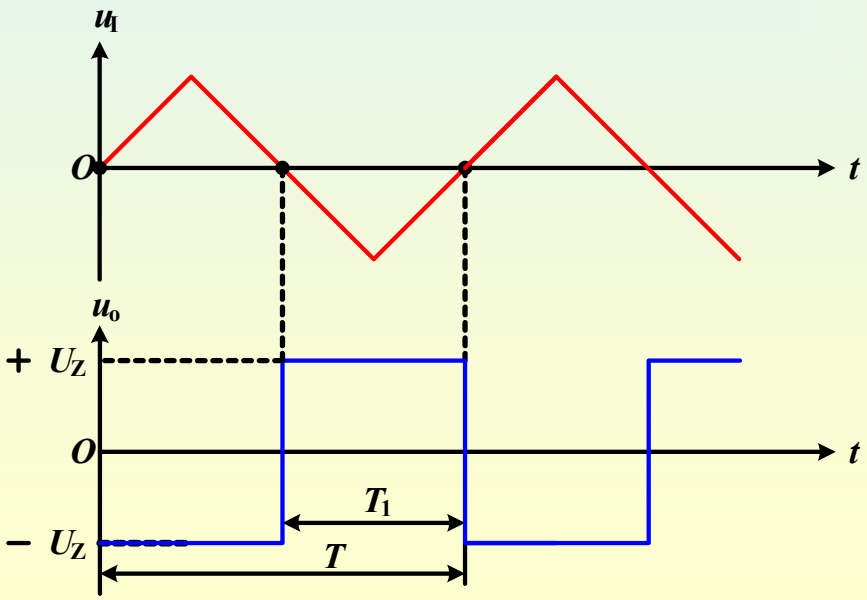
2) 电路的阈值电压$U_{TH} = U_{REF}$

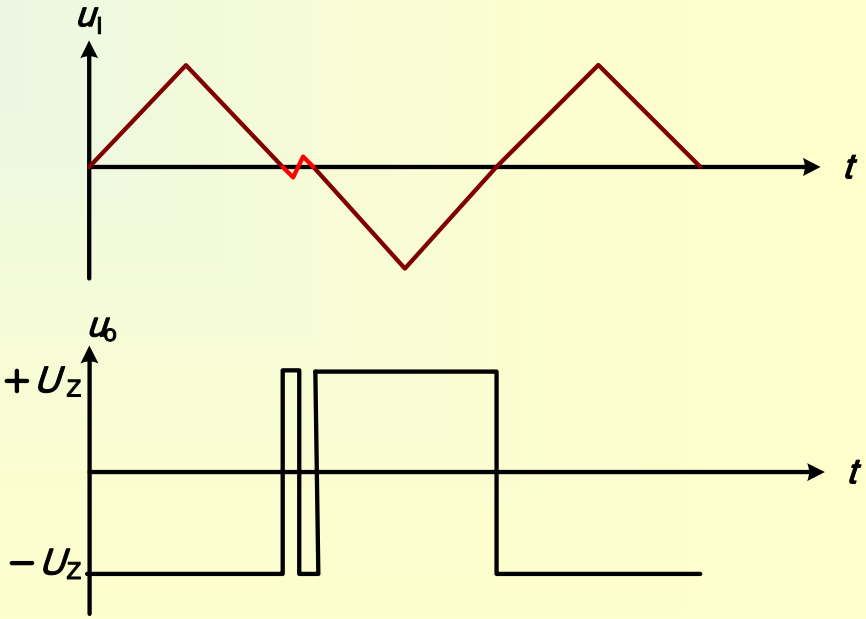
运放输入端并联的两个二极管作用是保护运放的输入级 。

运放输出端的R与D_Z构成输出限幅电路。

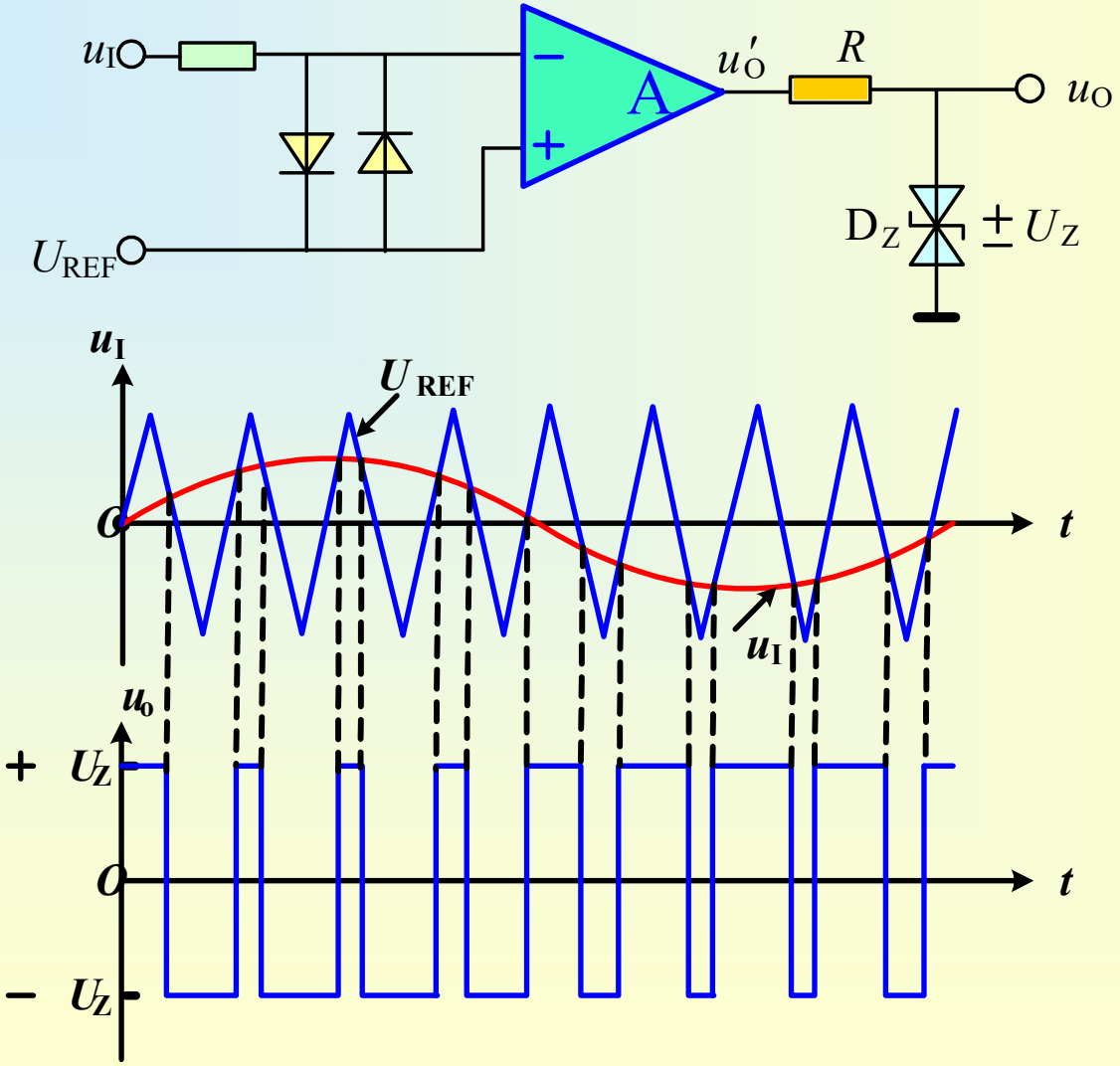
波形变换（波形整形，设$U_{REF}=0$）





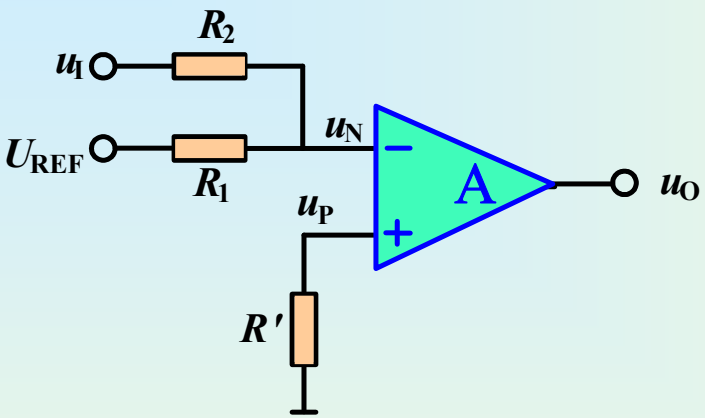


脉宽调制(PWM)

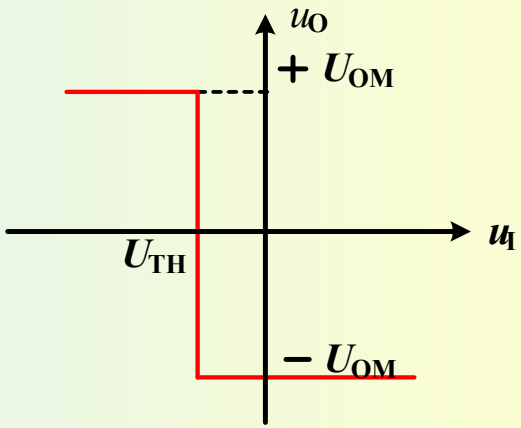


单限比较器输出的PWM波形

[例] 单限比较器的参考电压U_{REF}>0V，分析其电压传输特性。



(a)



(b)

解：根据“虚断”和叠加定理

$$u_P = 0 \qquad u_N = \frac{R_1}{R_1 + R_2} u_I + \frac{R_2}{R_1 + R_2} U_{REF}$$

令 $u_N = u_P = 0$ 求得 $U_{TH} = -\frac{R_2}{R_1} U_{REF}$

得出电路的电压传输特性。

5.3.3 滞回比较器

一、反相滞回比较器

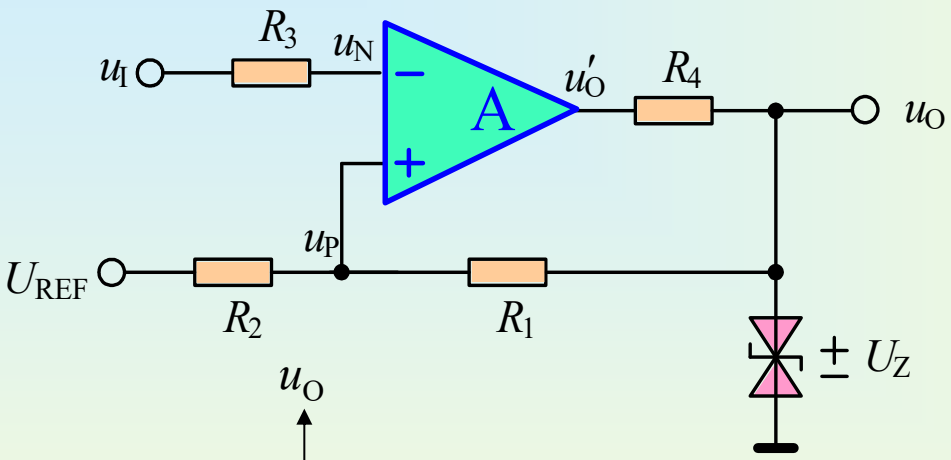
$$u_N = u_i, u_P = \frac{R_1}{R_1+R_2}U_{\mathrm{REF}} + \frac{R_2}{R_1+R_2}U_{\mathrm{O}}$$

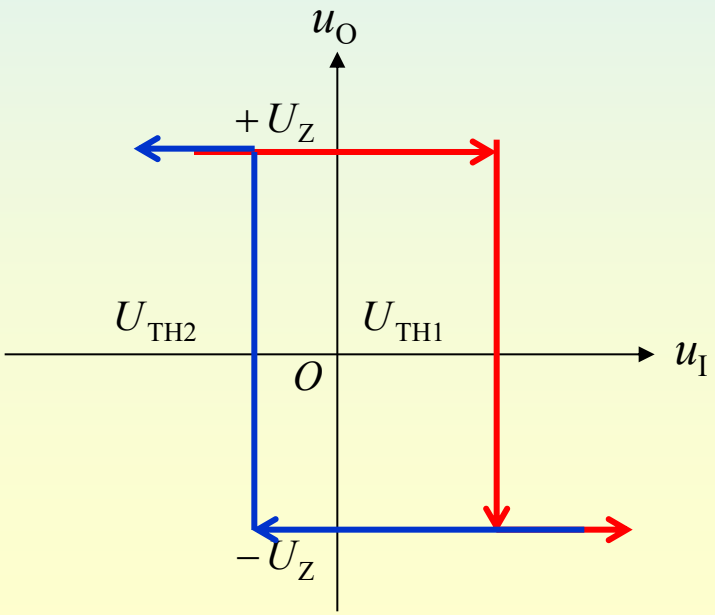
当$u_N = u_P$时，$u_i \to u_{TH}$

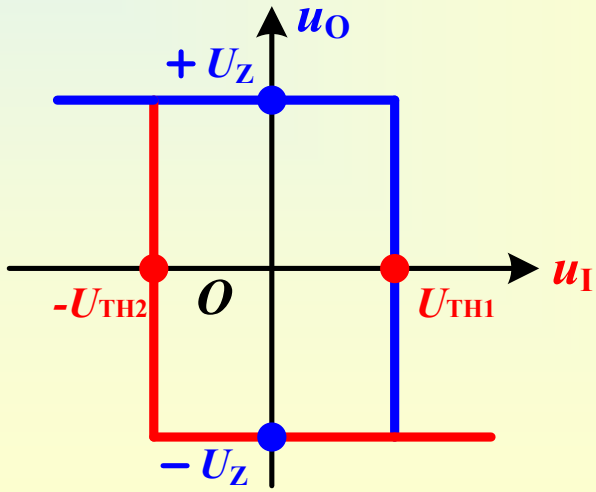
$$U_{\mathrm{TH1}} = \frac{R_1}{R_1+R_2}U_{\mathrm{REF}} + \frac{R_2}{R_1+R_2}U_{\mathrm{Z}}$$

$$U_{\mathrm{TH2}} = \frac{R_1}{R_1+R_2}U_{\mathrm{REF}} - \frac{R_2}{R_1+R_2}U_{\mathrm{Z}}$$

$$\Delta U = U_{\mathrm{TH1}} - U_{\mathrm{TH2}} = \frac{2R_2}{R_1+R_2}U_{\mathrm{Z}}$$



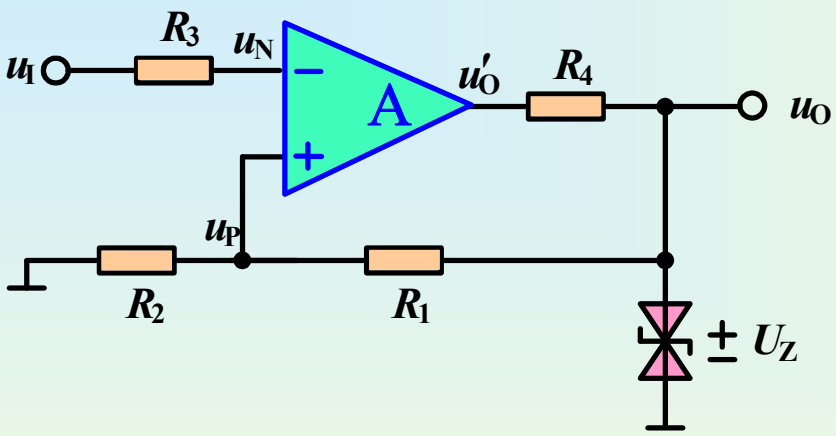


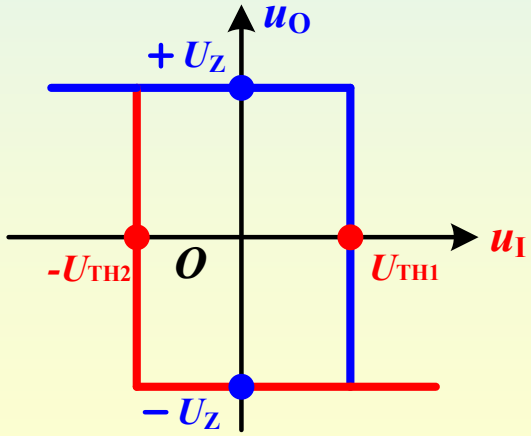


当U_{REF}=0时，U_{TH}=±$R_2U_Z/(R_1+R_2)$，

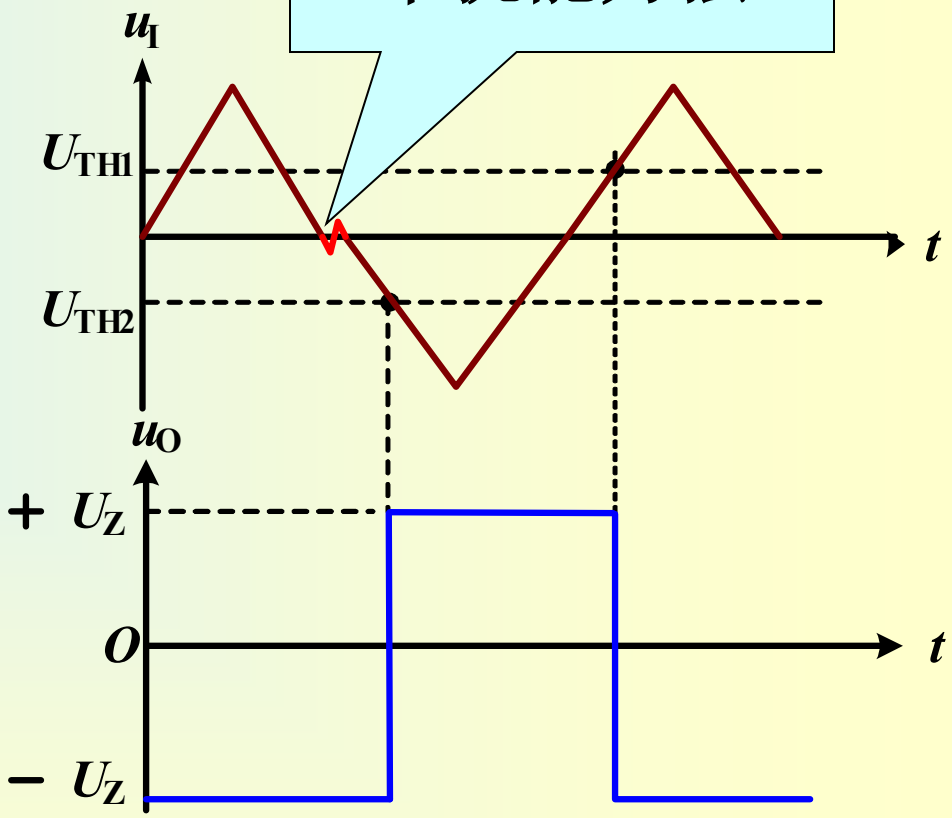
即$U_{TH2} = -U_{TH1}$。阈值电压纵轴对称。

滞回比较器抗干扰能力示意

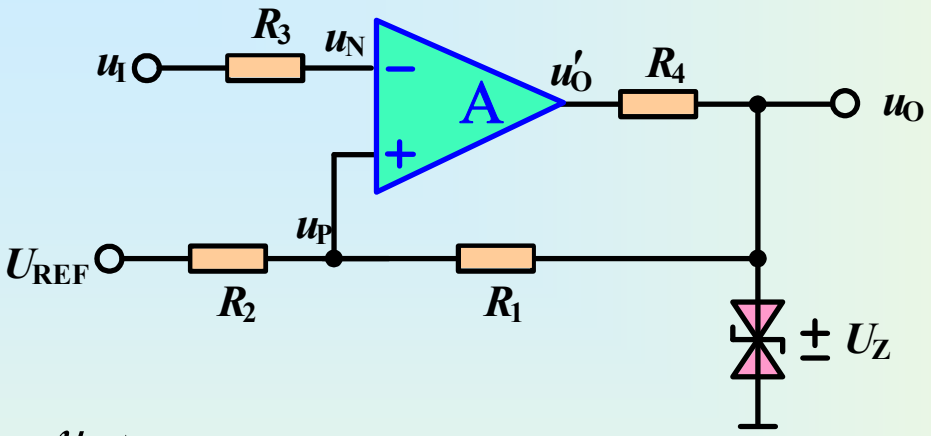


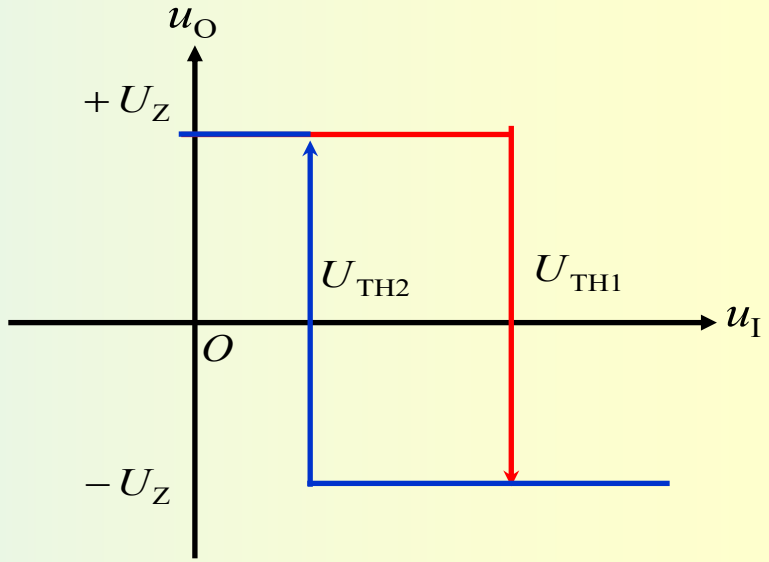


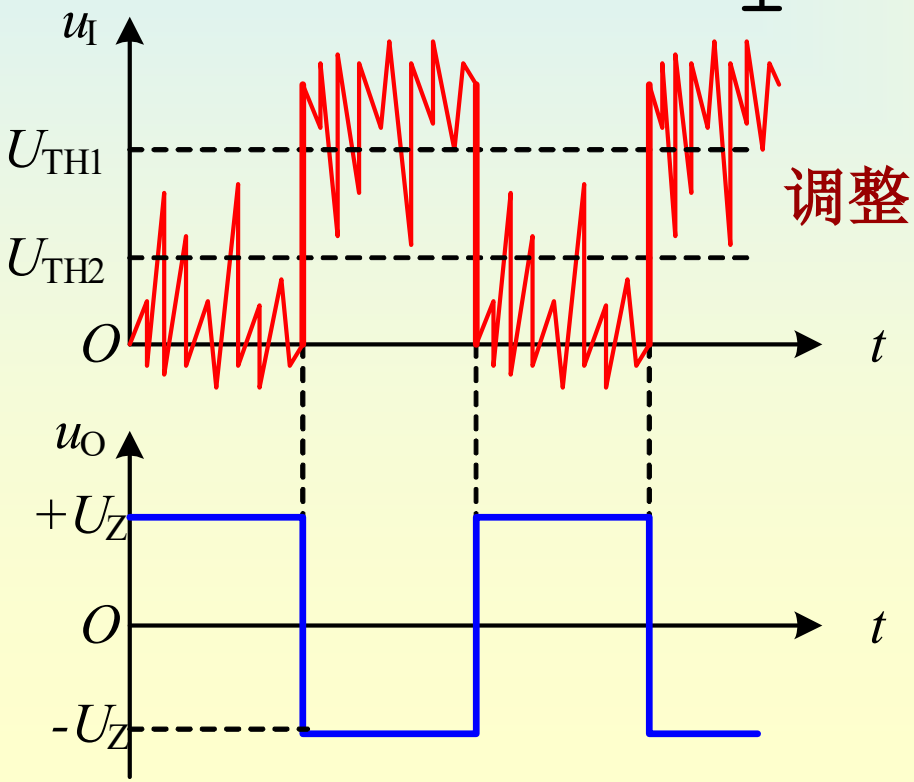
滞回比较器抗干扰能力强



滞回比较器抗干扰能力示意





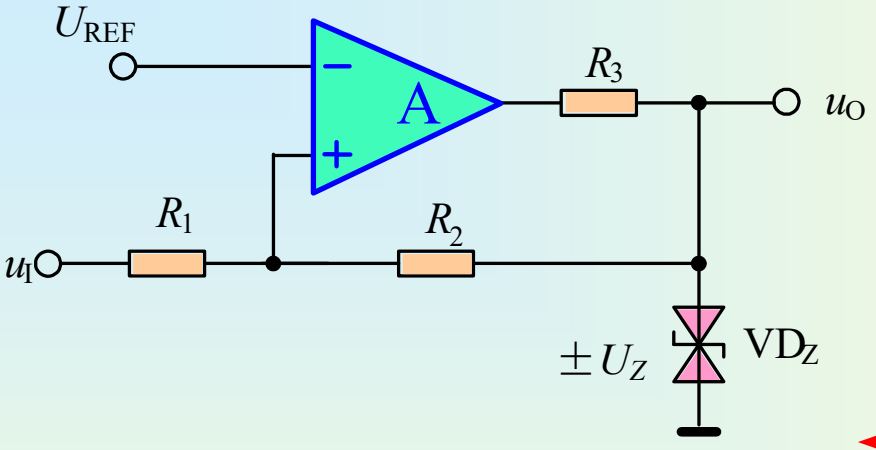


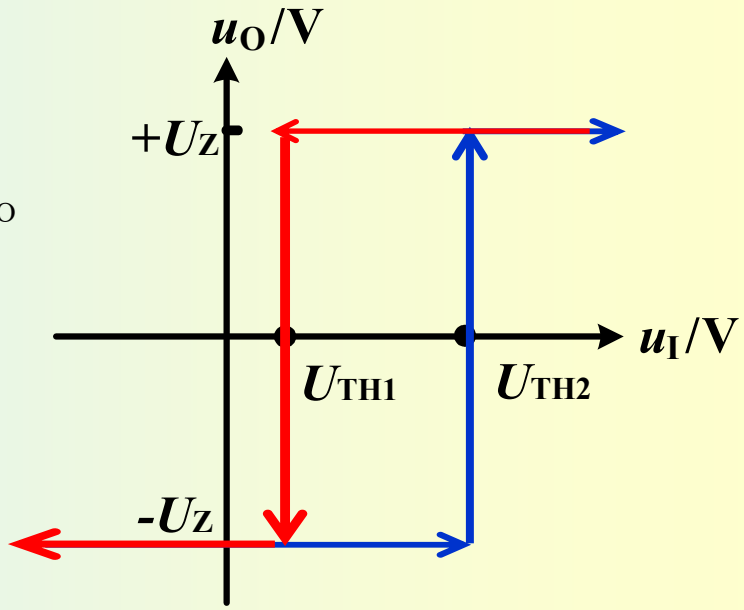
调整U_{REF}控制滞回电压ΔU的位置。

$$U_{TH1} = \frac{R_1}{R_1 + R_2} U_{REF} + \frac{R_2}{R_1 + R_2} U_Z$$

$$U_{TH2} = \frac{R_1}{R_1 + R_2} U_{REF} - \frac{R_2}{R_1 + R_2} U_Z$$

二、同相滞回比较器





根据“虚断”和叠加定理

$$u_P = \frac{R_2}{R_1+R_2}U_I + \frac{R_1}{R_1+R_2}u_O$$

$$u_N = U_{REF}$$

当$u_N = u_P$时，$u_i \rightarrow u_{TH}$

$$U_{TH} = \frac{R_1+R_2}{R_2}U_{REF} - \frac{R_1}{R_2}u_O$$

$$U_{TH1} = \frac{R_1+R_2}{R_2}U_{REF} - \frac{R_1}{R_2}U_Z$$

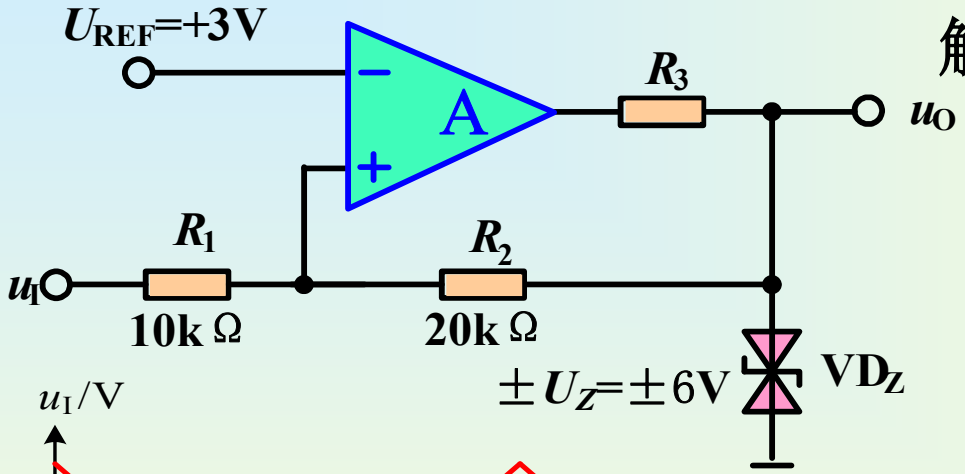
$$U_{TH2} = \frac{R_1+R_2}{R_2}U_{REF} + \frac{R_1}{R_2}U_Z$$

$$\Delta U = U_{TH2} - U_{TH1} = 2\frac{R_1}{R_2}U_Z$$

[例] 设运放的最大输出电压±U_{OM} = ±12V，试求：

1. 分析电路的电压传输特性；

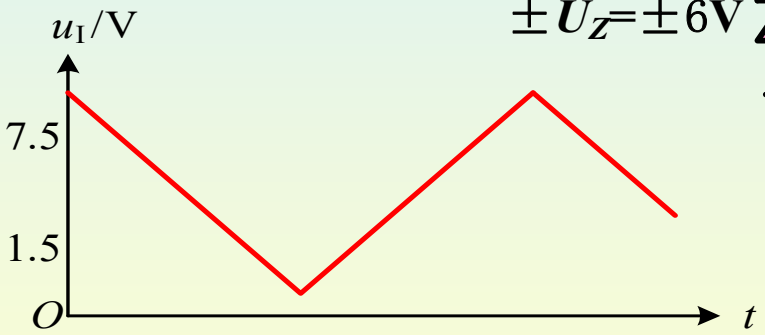
2. 根据电路输入电压信号画出电路的输出电压波形。

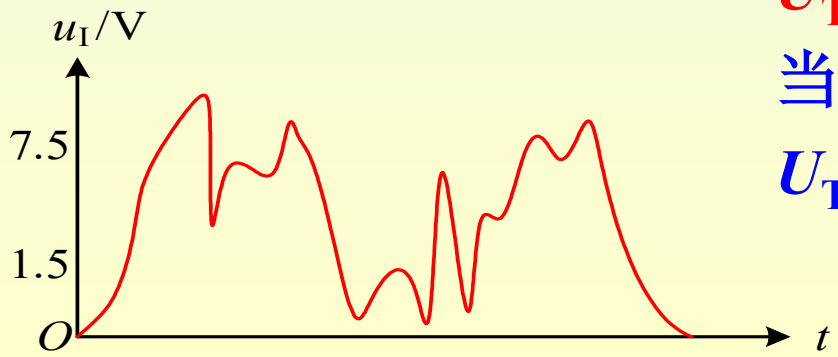


解：

$$U_{TH1}=\frac{R_1+R_2}{R_2}U_{REF}-\frac{R_1}{R_2}U_Z$$

$$U_{TH2}=\frac{R_1+R_2}{R_2}U_{REF}+\frac{R_1}{R_2}U_Z$$



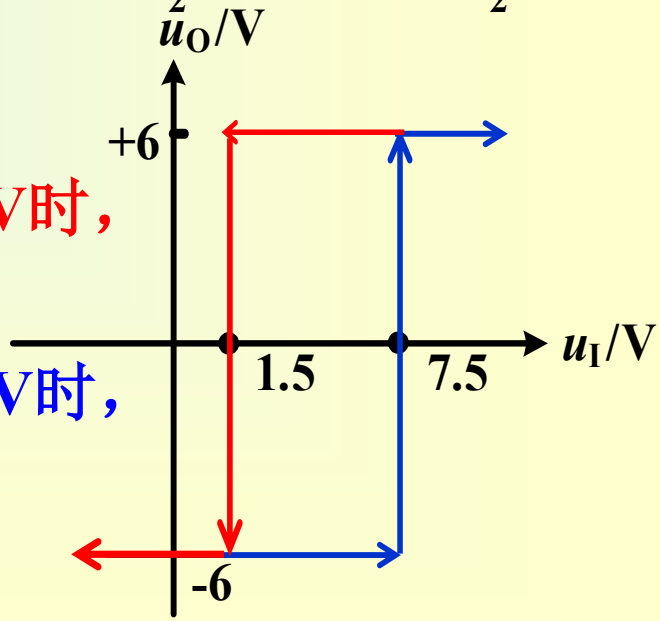


当u_O=U_Z=+6V时，

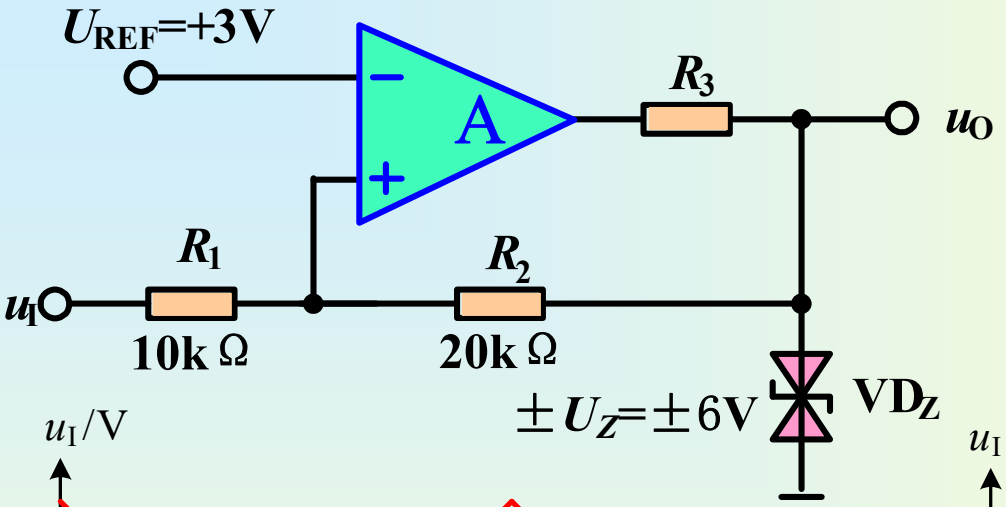
U_{TH1}=1.5V；

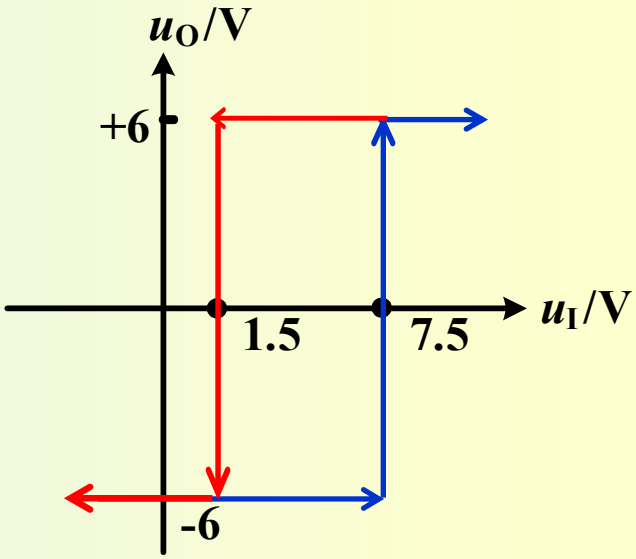
当u_O=-U_Z=-6V时，

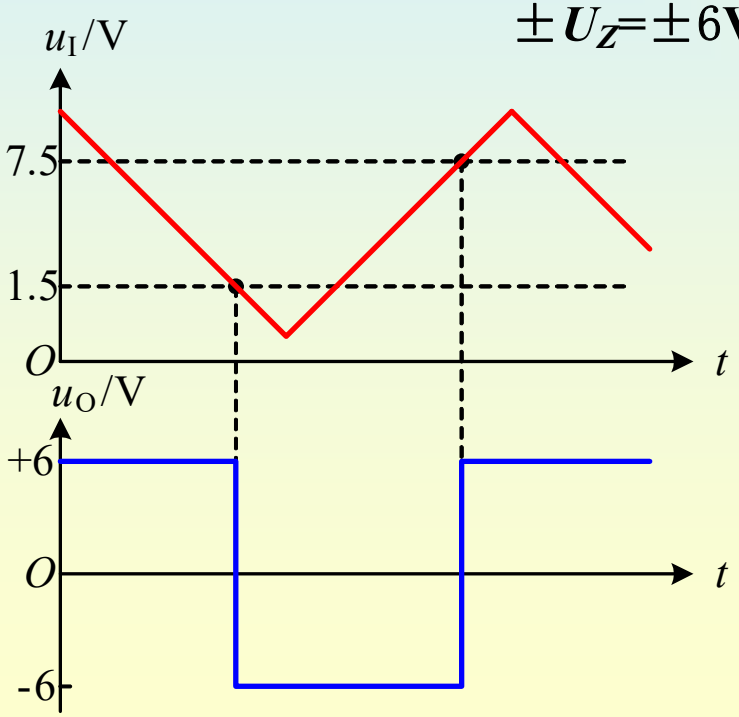
U_{TH2}=7.5V。

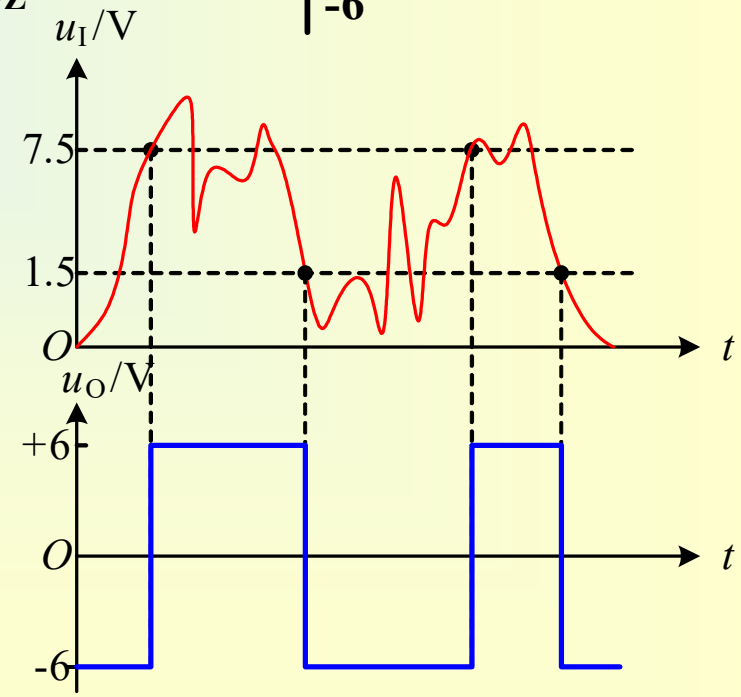


[例解]

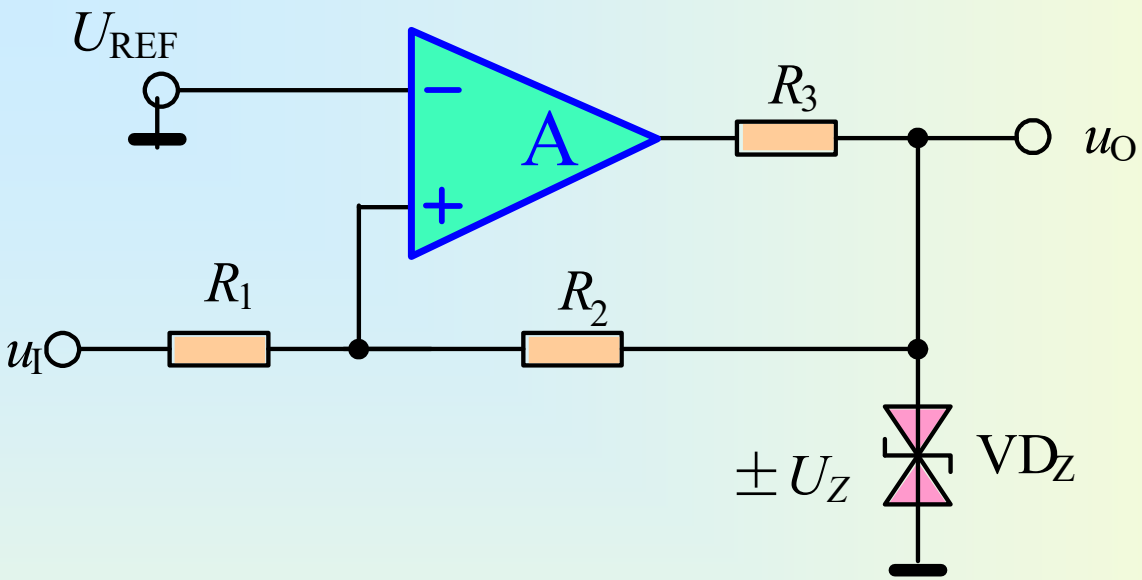






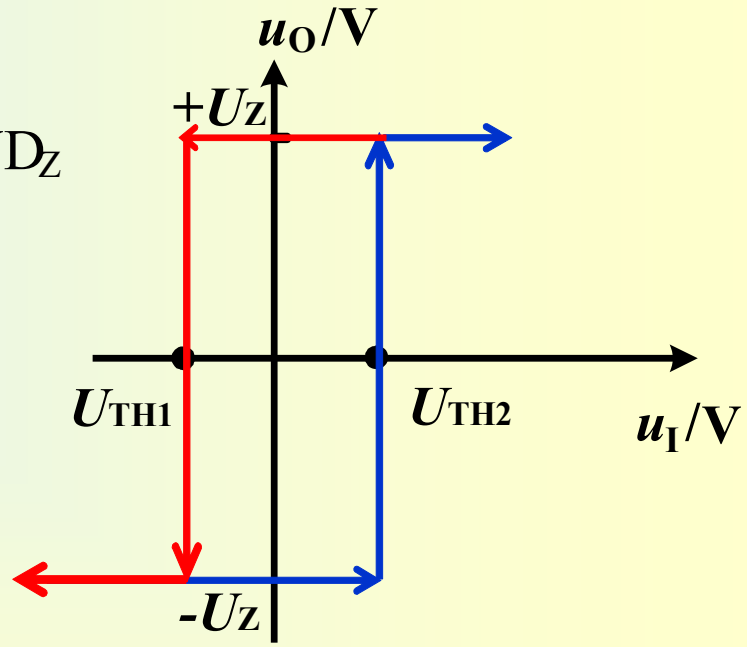


输入和输出电压波形



$$U_{TH1}=\frac{R_1+R_2}{R_2}U_{REF}-\frac{R_1}{R_2}U_Z$$

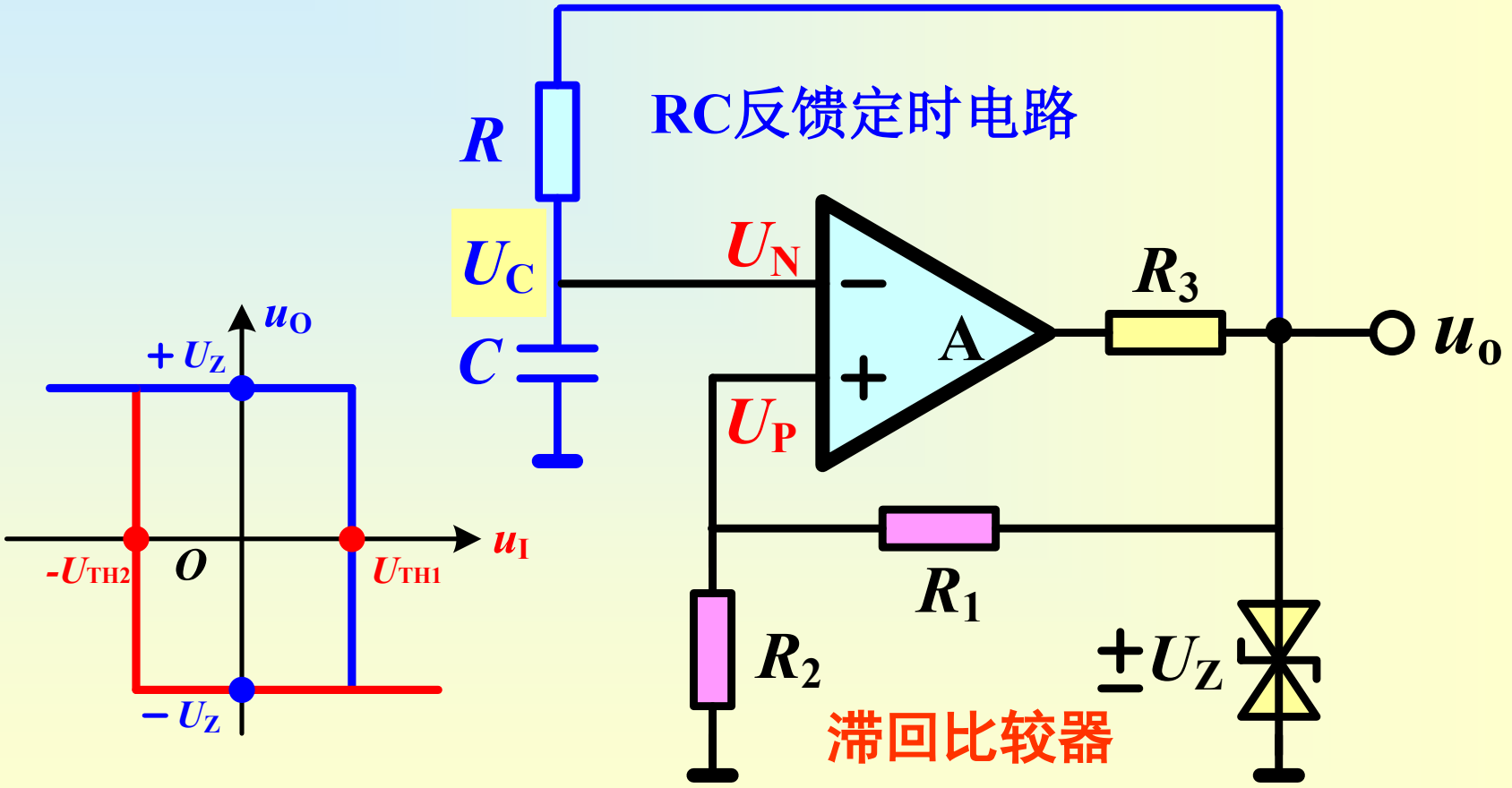
$$U_{TH2}=\frac{R_1+R_2}{R_2}U_{REF}+\frac{R_1}{R_2}U_Z$$



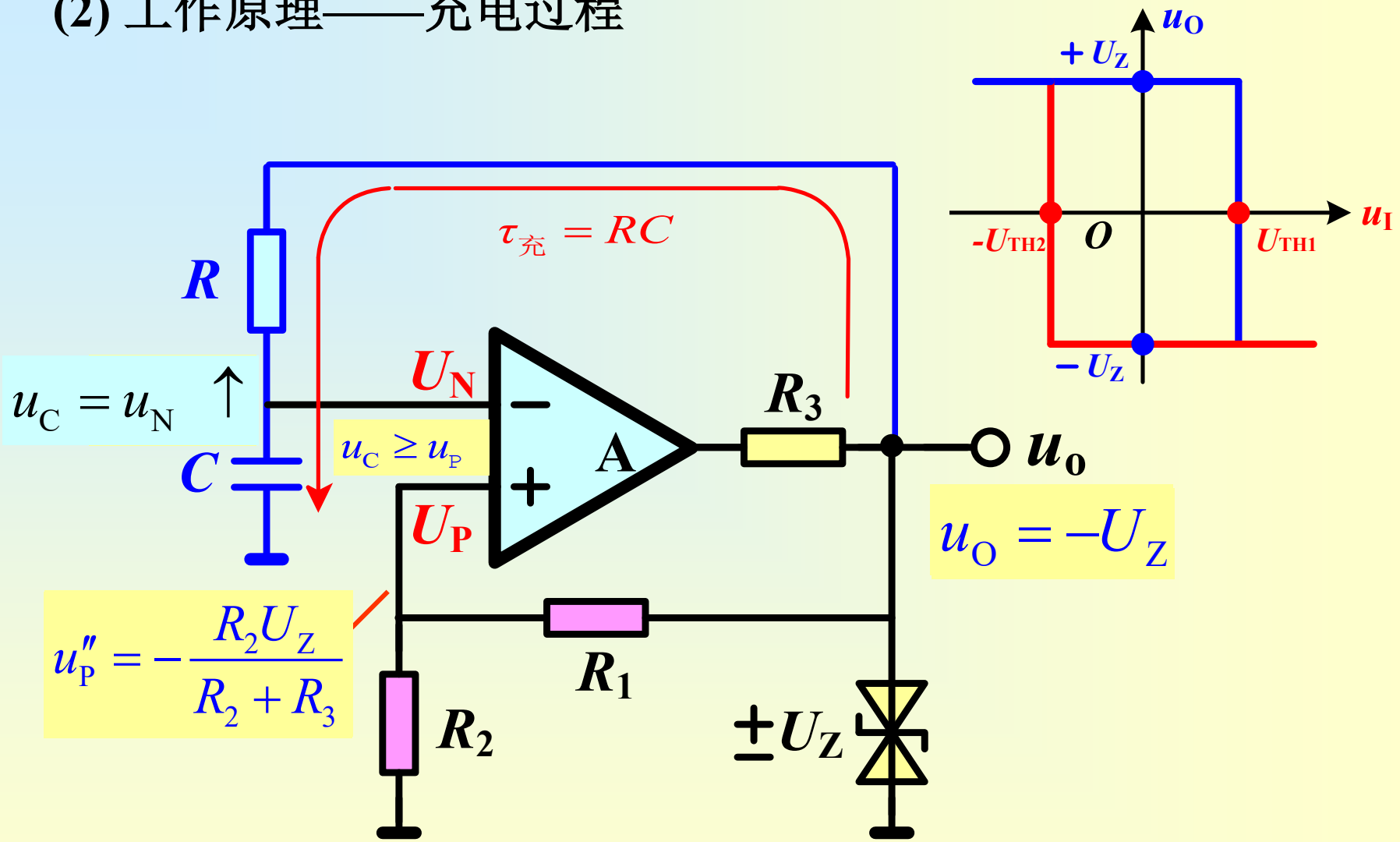
当U_{REF}=0时，$U_{TH}=\pm U_Z R_1/R_2$，即$U_{TH1}=-U_{TH2}$。阈值电压纵轴对称，回差不变$\Delta U=2U_Z R_1/R_2$。

5.3.4 矩形波发生电路（反相滞回比较器构建）

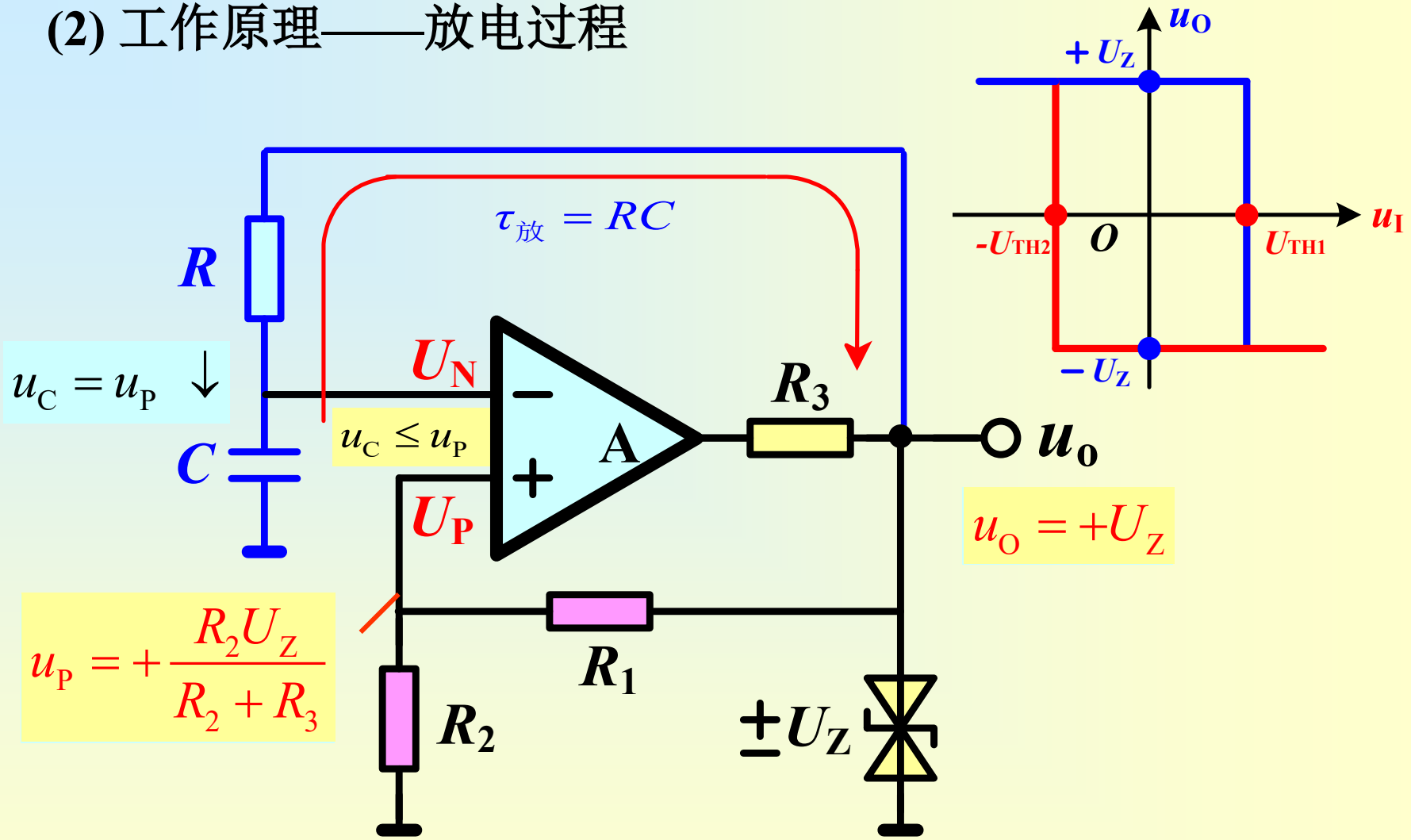
(1) 电路组成



(2) 工作原理——充电过程

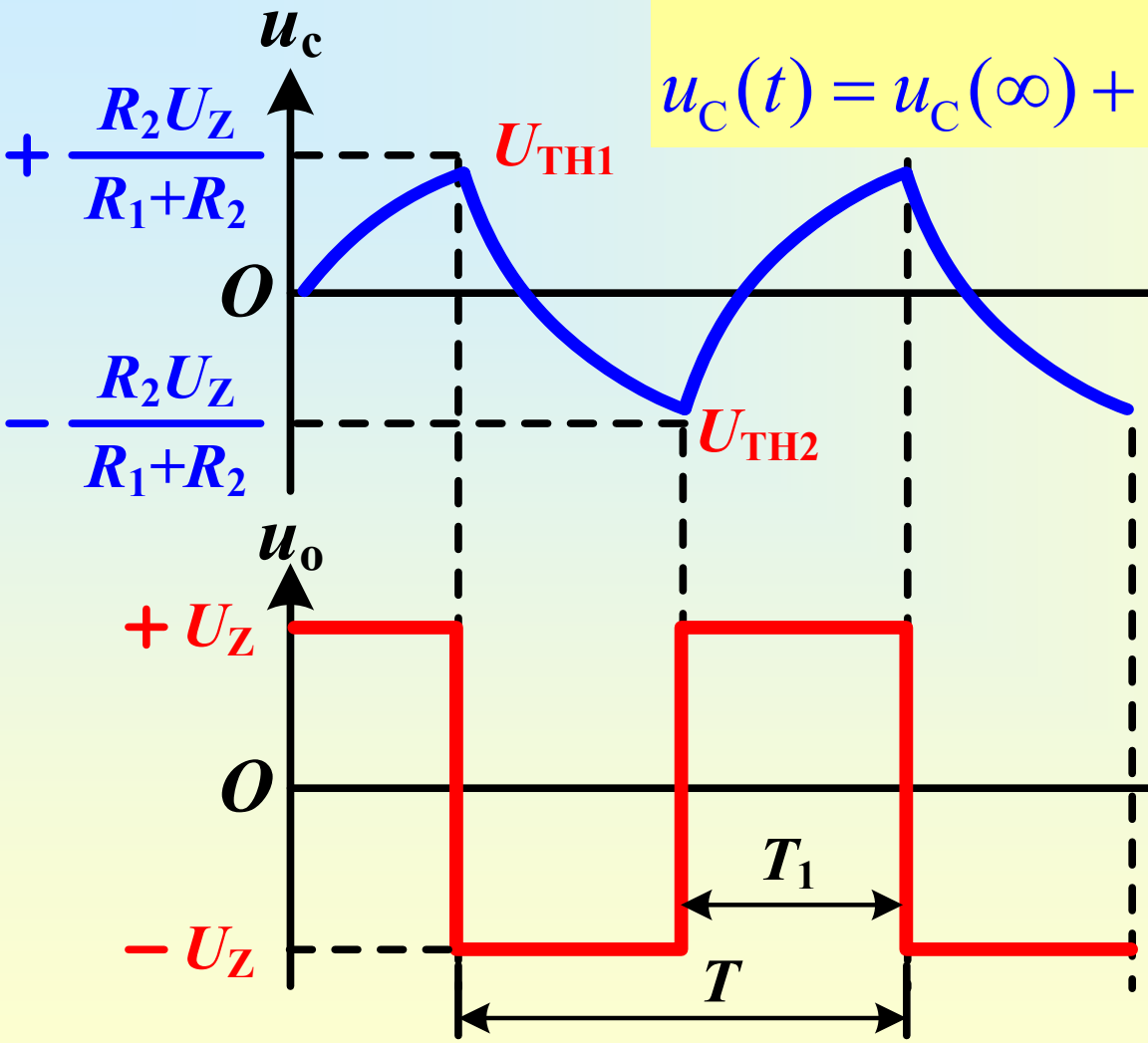

u_O
$+U_Z$
$-U_{TH2}$
O
U_{TH1}
u_I
$-U_Z$
$\tau_{充} = RC$
R
C
$u_C = u_N$
U_N
U_P
A
R_3
$u_C \geq u_P$
u_O
$u_O = -U_Z$
$u''_P = -\dfrac{R_2 U_Z}{R_2 + R_3}$
R_1
R_2
$\pm U_Z$

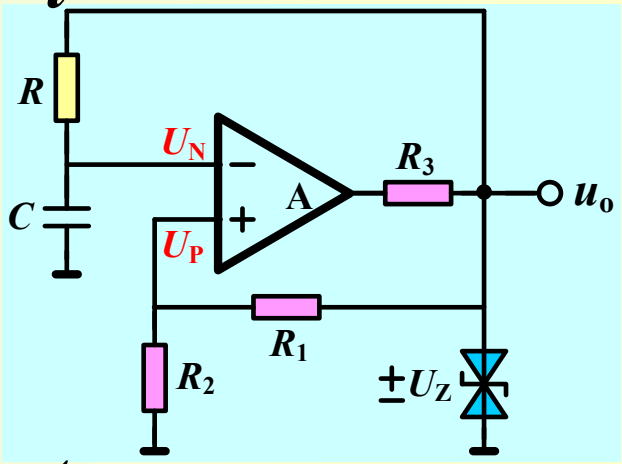
(2) 工作原理——放电过程


u_O
$+U_Z$
$-U_{TH2}$
O
U_{TH1}
u_I
$-U_Z$
$\tau_{放} = RC$
R
C
$u_C = u_P$ ↓
U_N
U_P
A
R_3
$u_C \le u_P$
u_O
$u_O = +U_Z$
$u_P = +\dfrac{R_2 U_Z}{R_2 + R_3}$
R_1
R_2
$\pm U_Z$

(3) 波形分析与参数计算

$$u_C(t) = u_C(\infty) + [u_C(0) - u_C(\infty)]e^{-\frac{t}{\tau}}$$

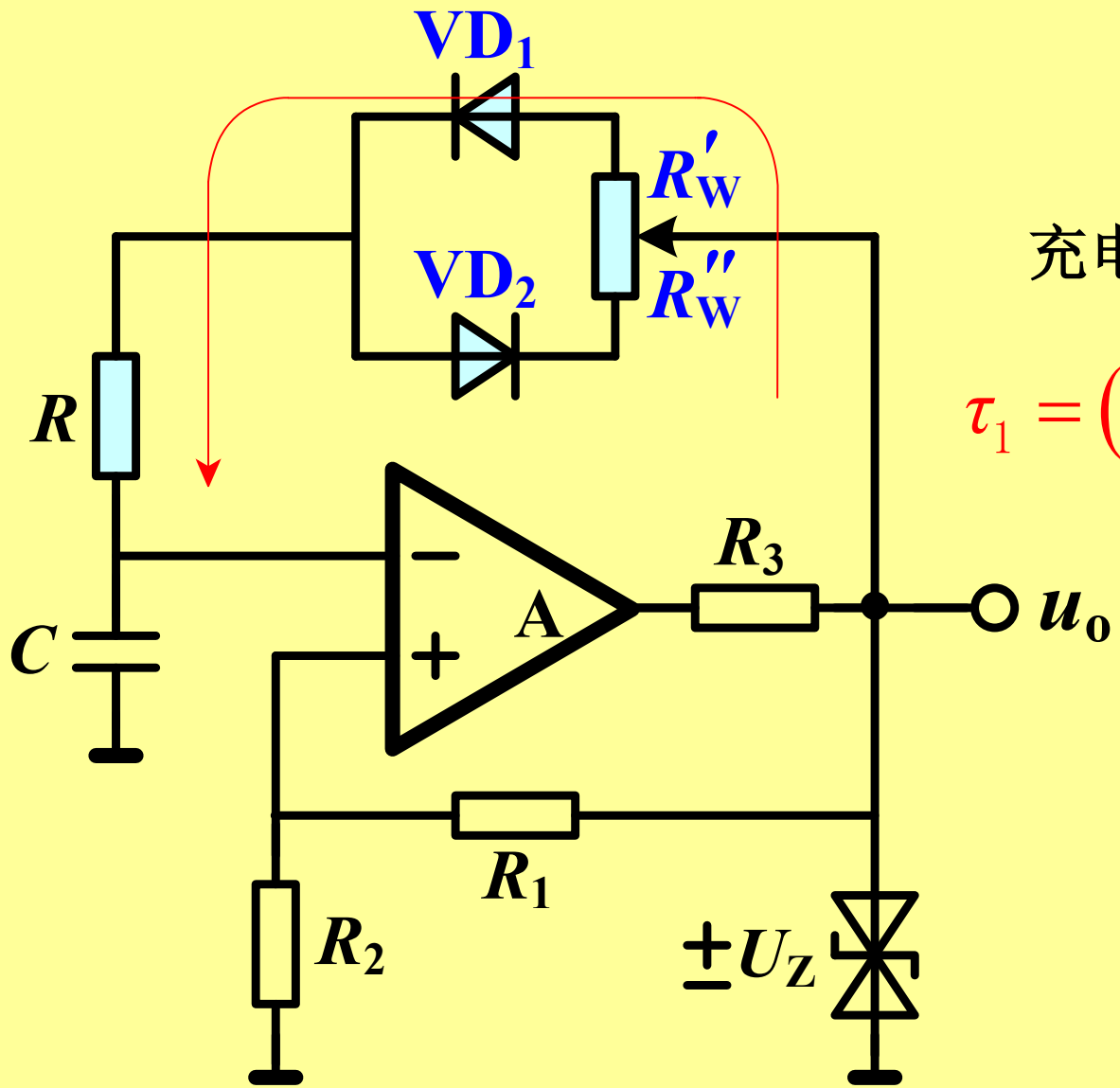




$$T = 2RC\ln(1+\frac{2R_2}{R_1})$$

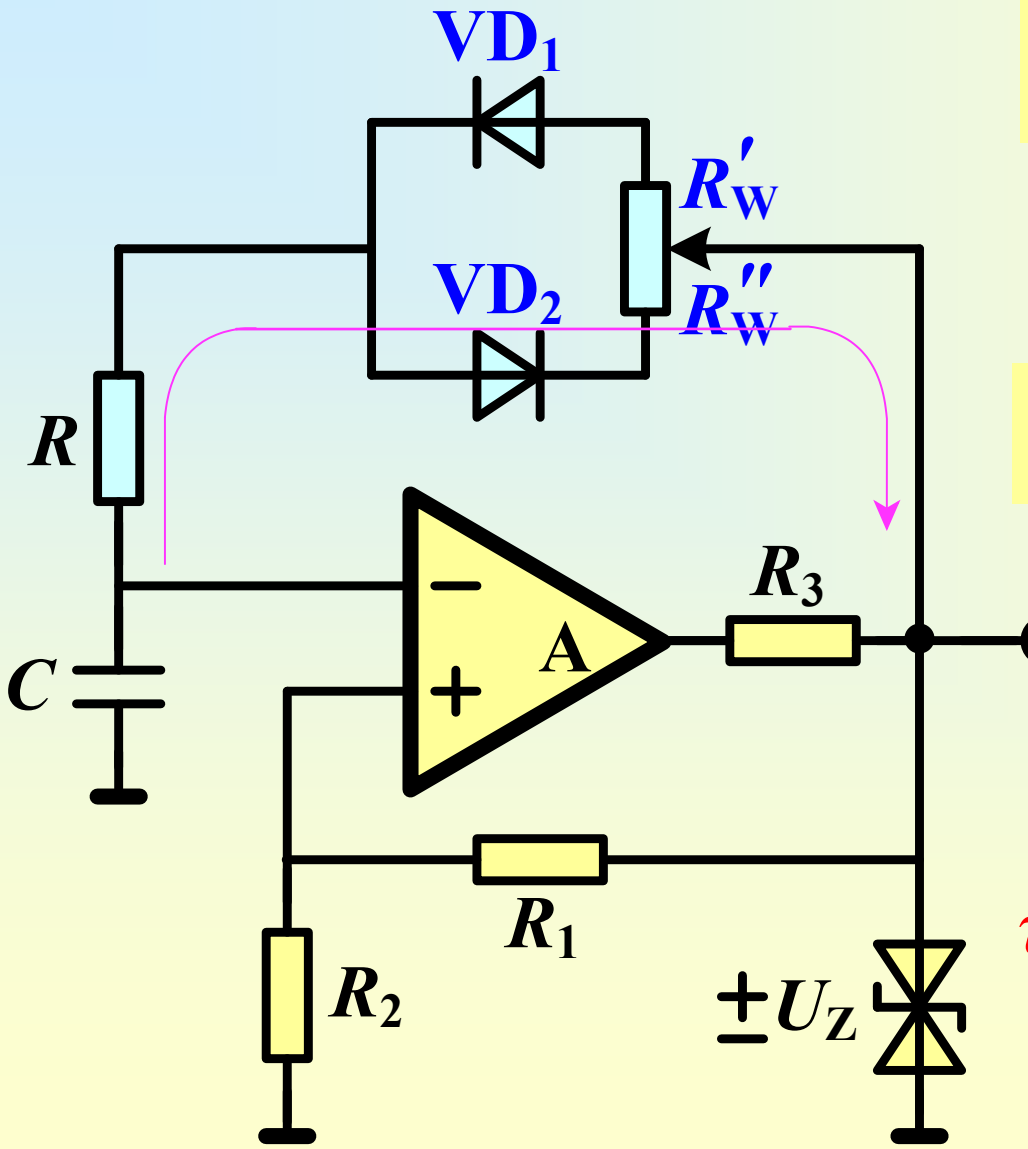
$$\text{占空比 } D = \frac{T_1}{T} \times 100\% = 50\%$$

讨论：若T和U_{om}不变，如何改变u_o的占空比？



充电时间常数

$$\tau_1 = \left(R + r_{d1} + R'_W\right)C$$



$$D=\frac{\tau_1}{\tau_1+\tau_2}\times 100\%$$

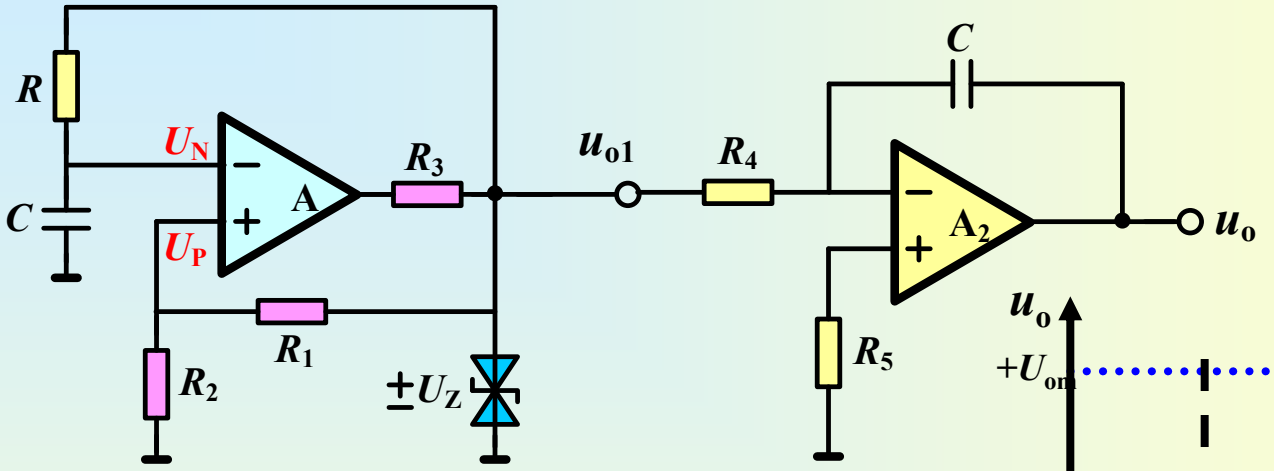
充电时间常数

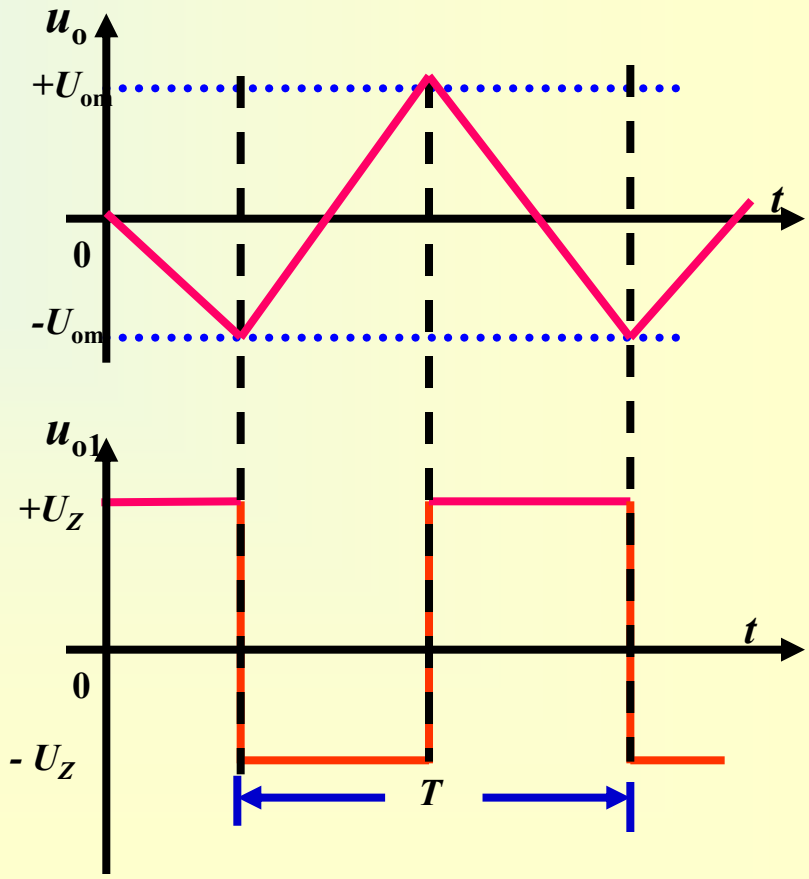
$$\tau_1=\left(R+r_{d1}+R'_W\right)C$$

放电时间常数

$$\tau_2=\left(R+r_{d2}+R''_W\right)C$$

5.3.5 三角波发生电路（同相滞回比较器构建）

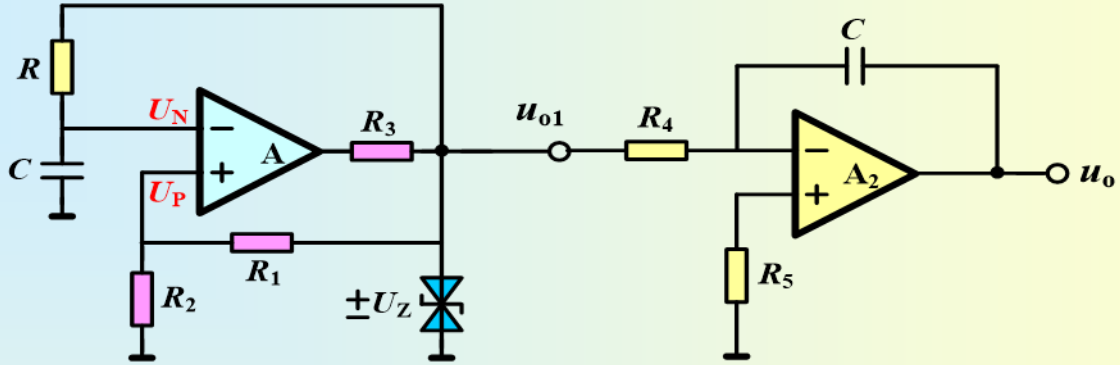


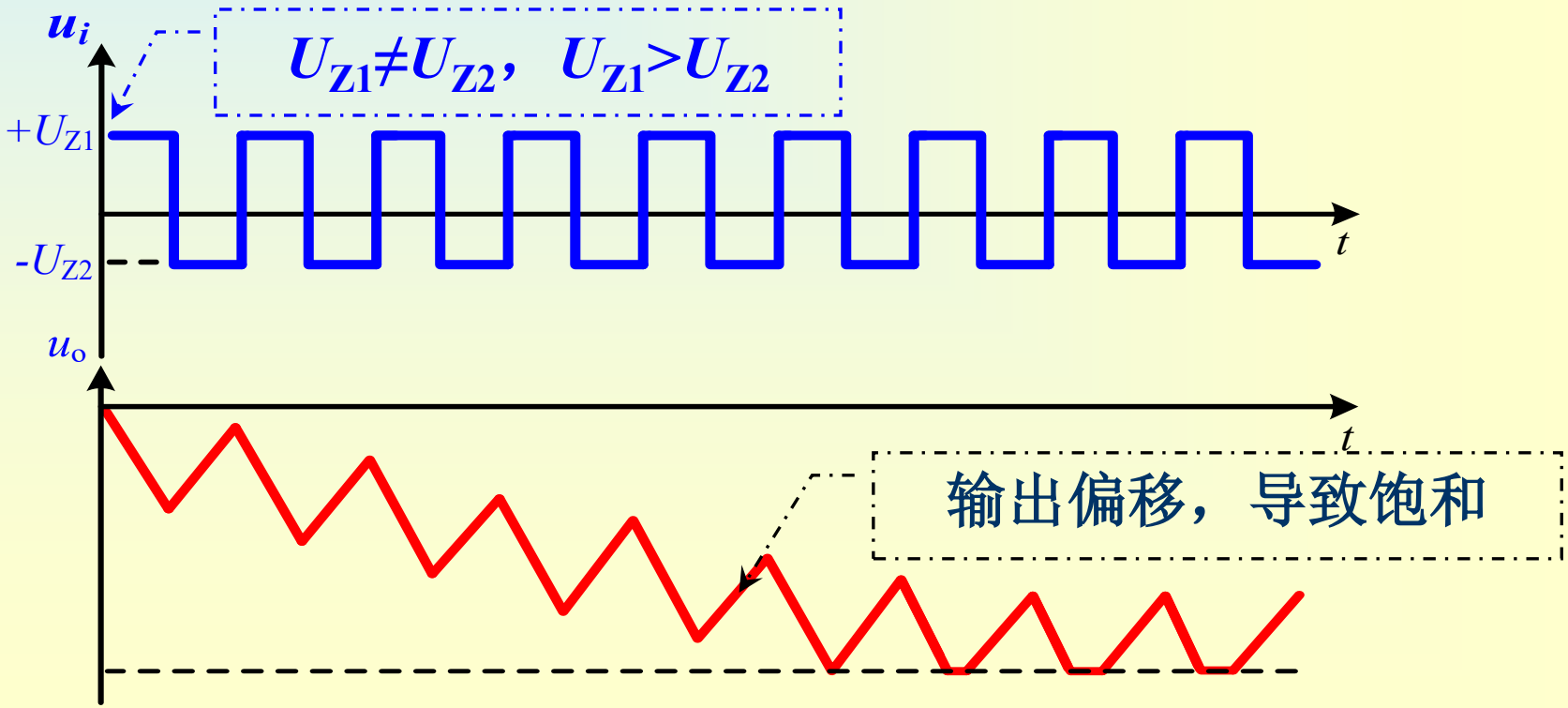


1、矩形波+积分电路

产生三角波？ ×

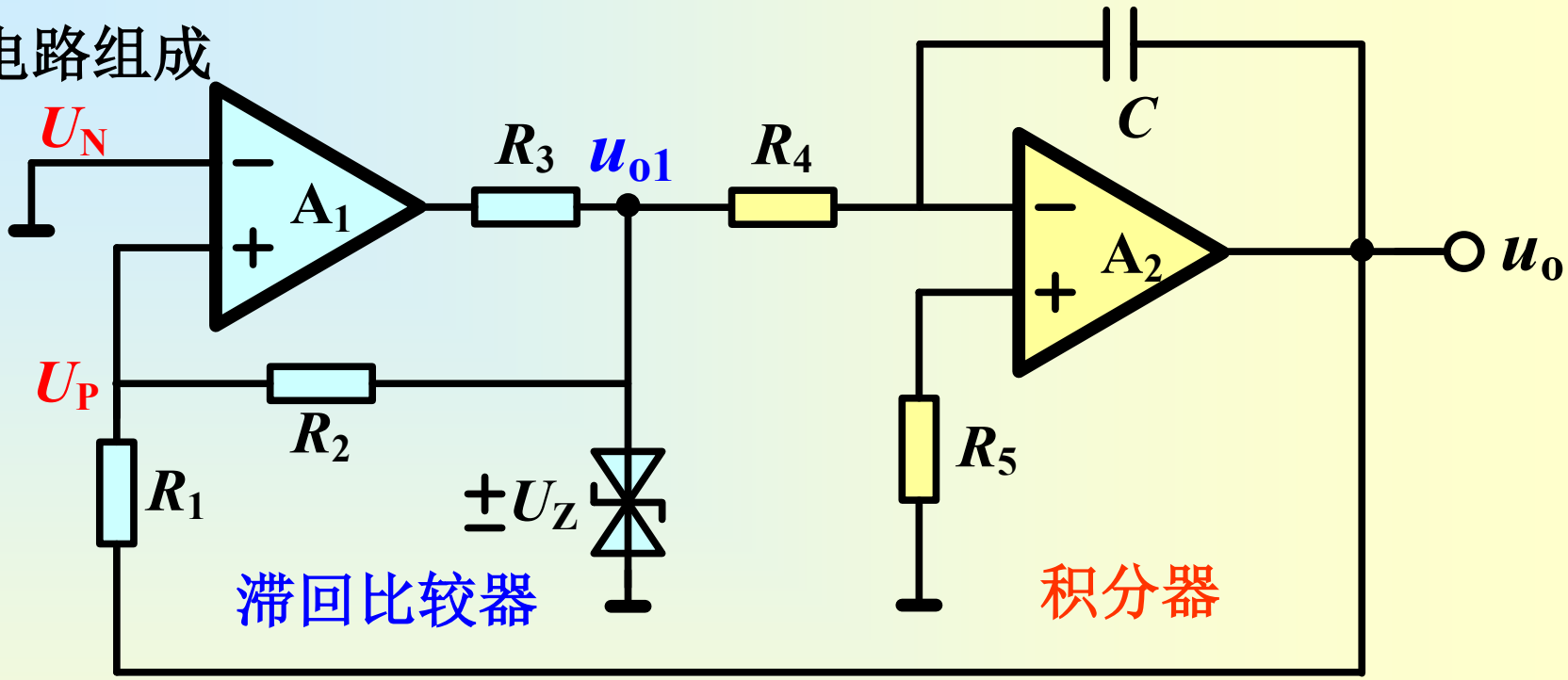
为什么不用矩形波+积分电路产生三角波?

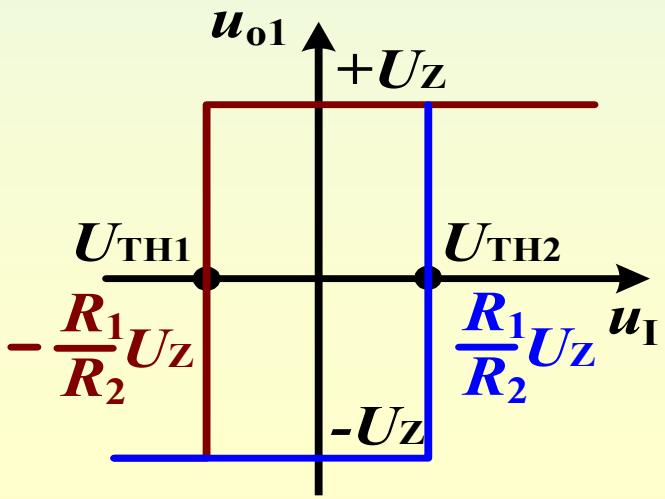




2 同相滞回比较器构建→三角波发生电路

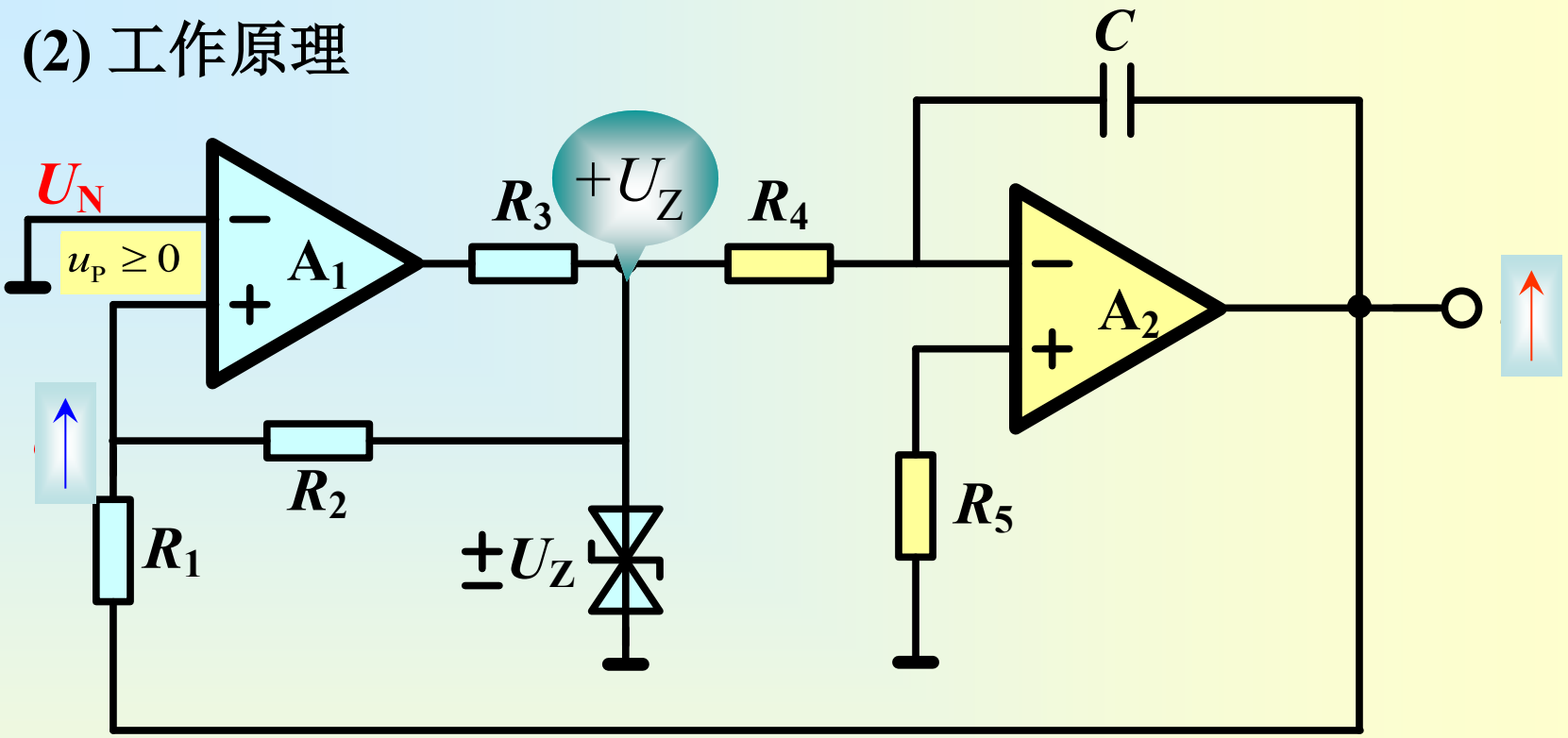
(1) 电路组成



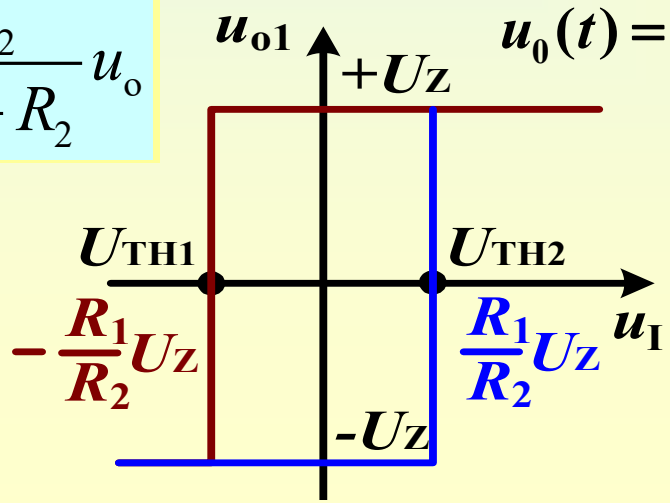


$$u_0(t) = -\frac{1}{R_4 C}\int_{t_1}^{t} u_{01}dt + u_o(t_1)$$

(2) 工作原理

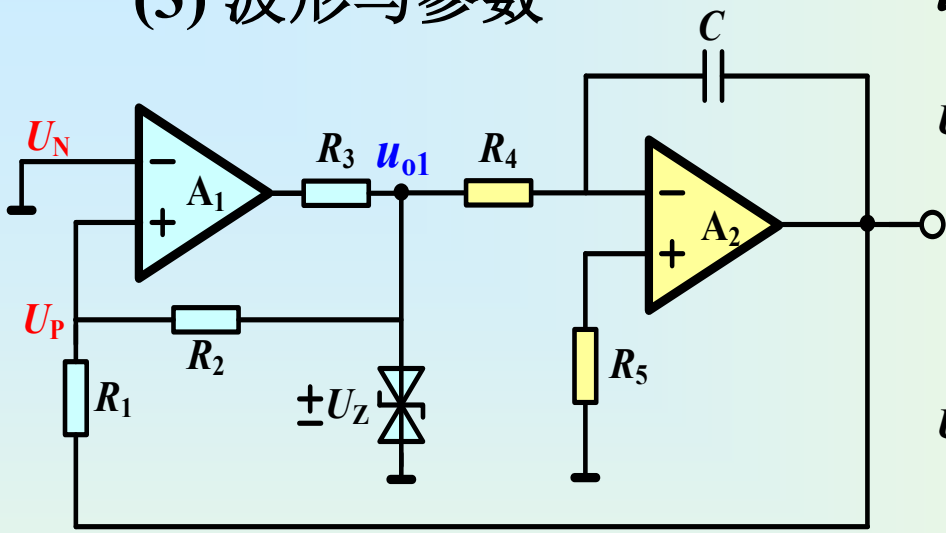


$$u_P = -\frac{R_1}{R_1+R_2}U_Z + \frac{R_2}{R_1+R_2}u_o$$

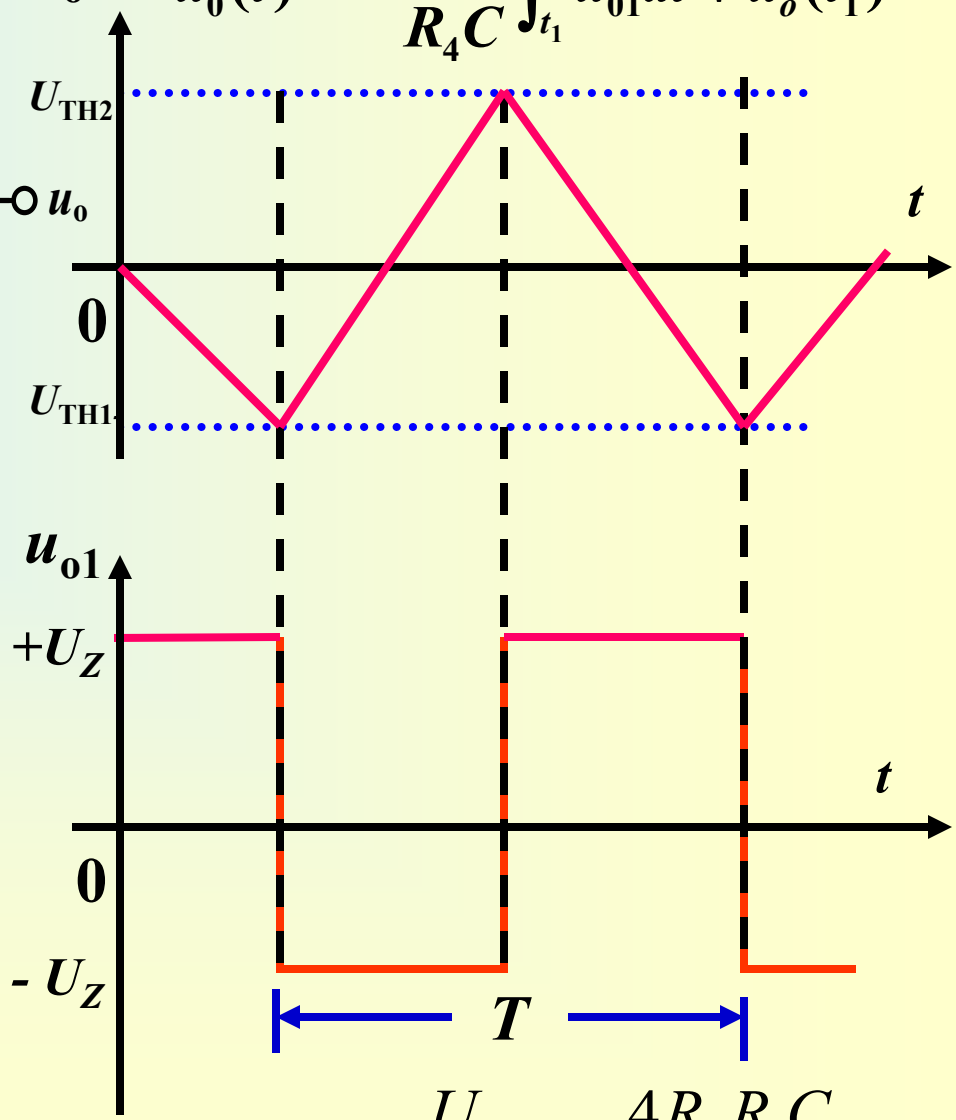


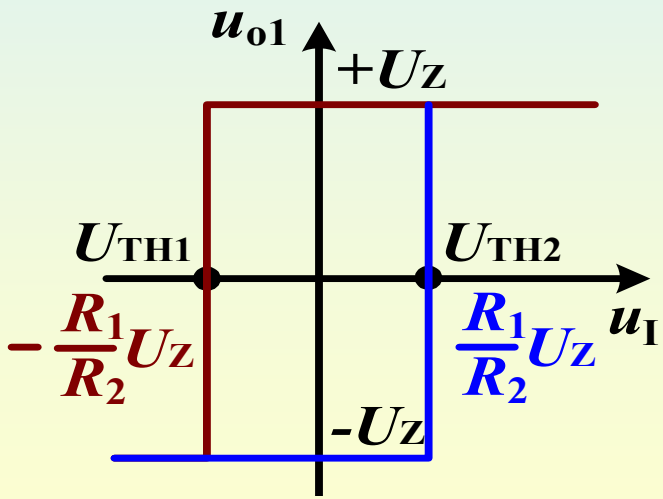
$$u_0(t) = -\frac{1}{R_4 C}\int_{t_1}^{t} u_{01}dt + u_o(t_1)$$

(3) 波形与参数



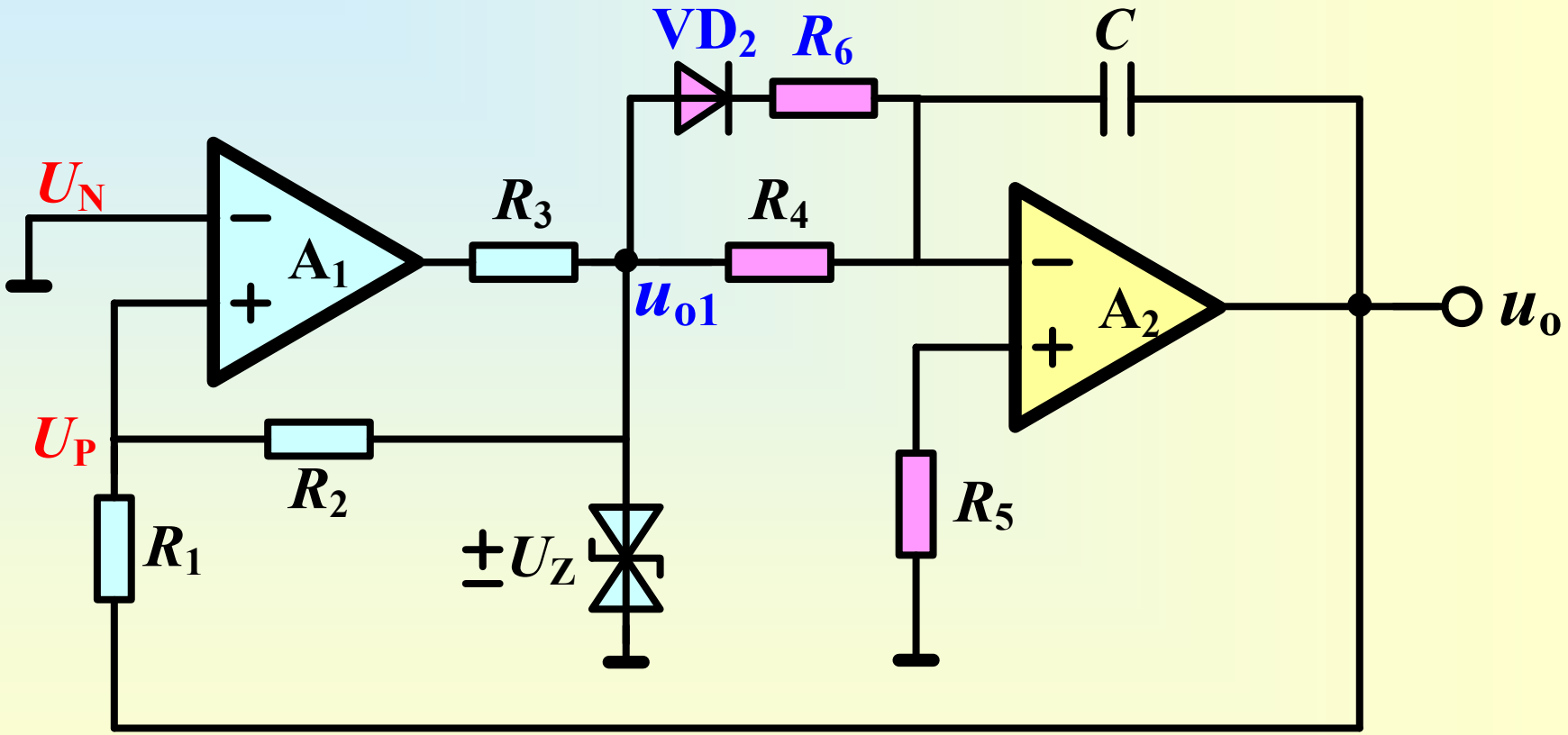
$$u_0(t) = -\frac{1}{R_4 C}\int_{t_1}^{t} u_{01}dt + u_o(t_1)$$

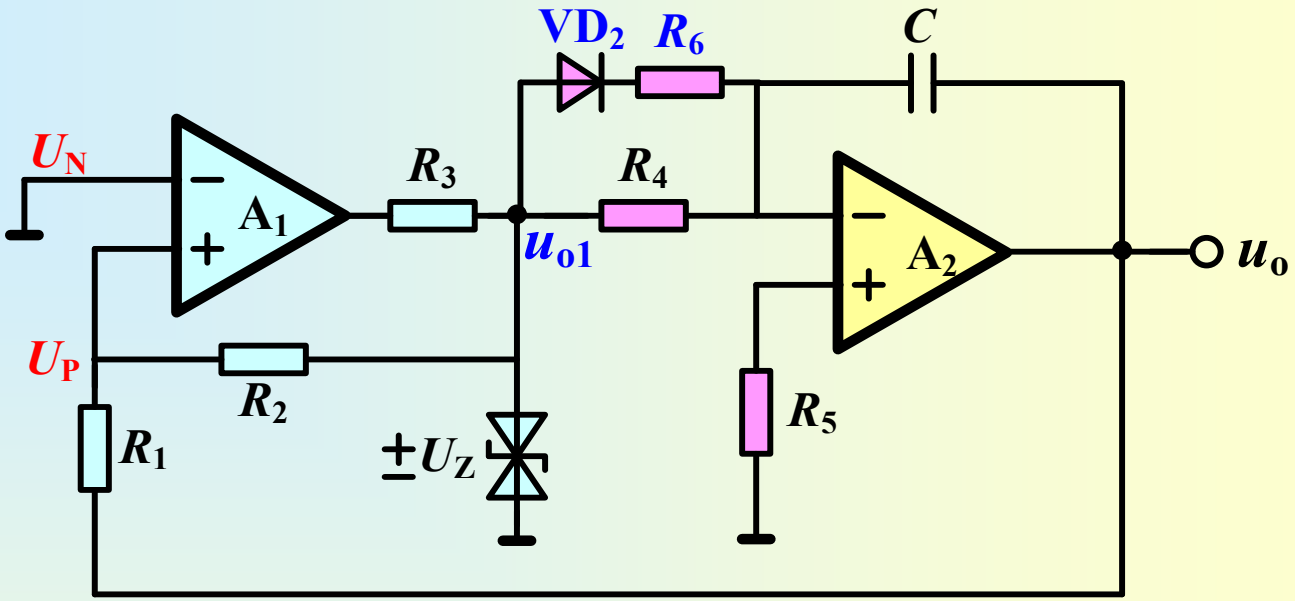


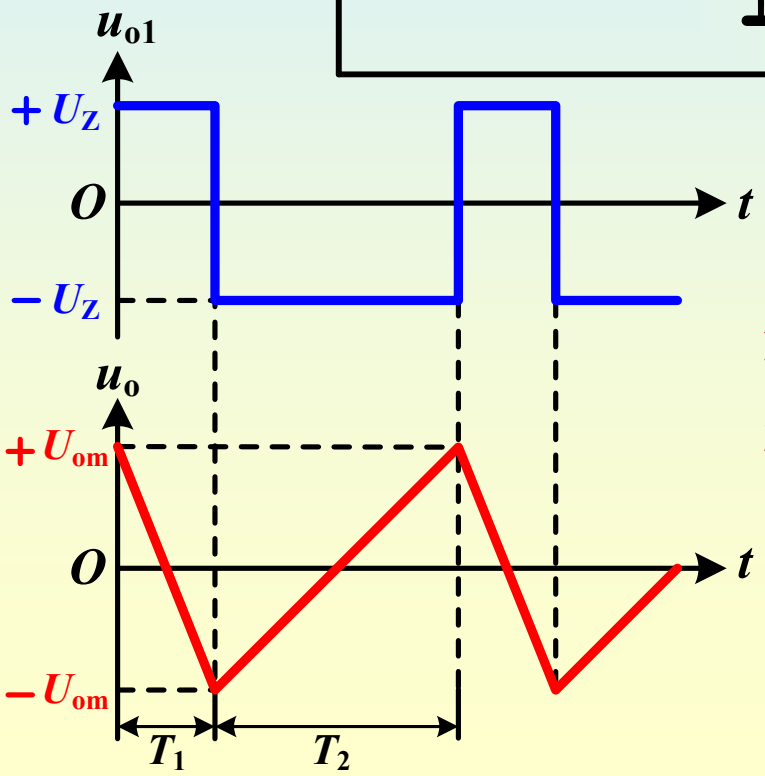


$$\pm U_{om} = \frac{R_1}{R_2}U_Z \longrightarrow T = 4R_4 C\frac{U_{om}}{U_Z} = \frac{4R_4 R_1 C}{R_2}$$

讨论：改变u_o的占空比？

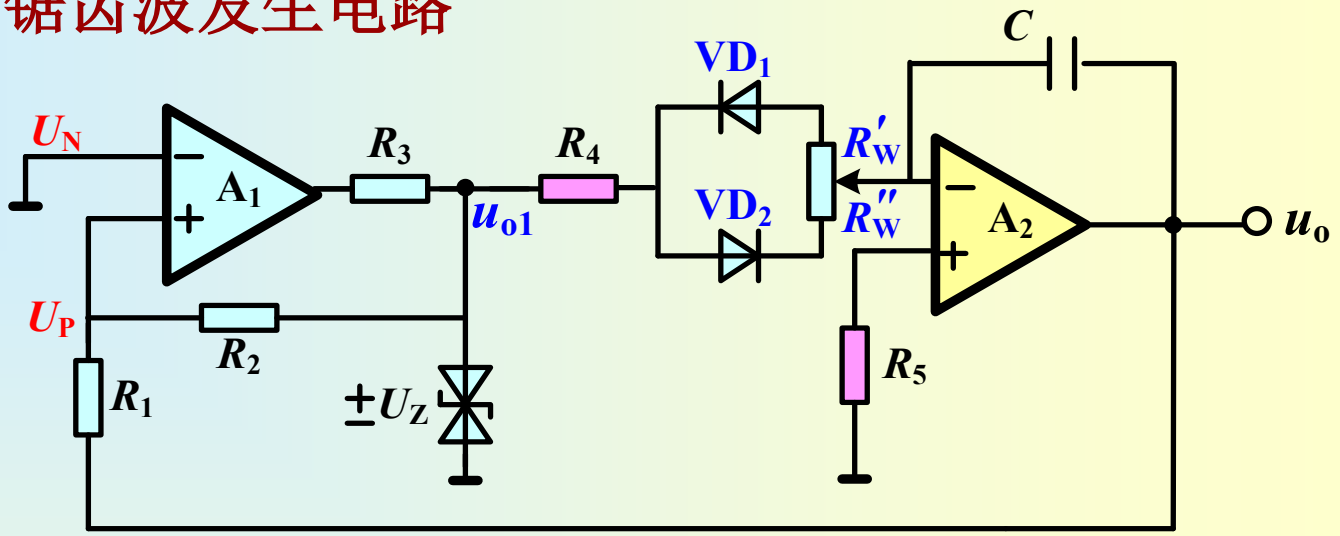


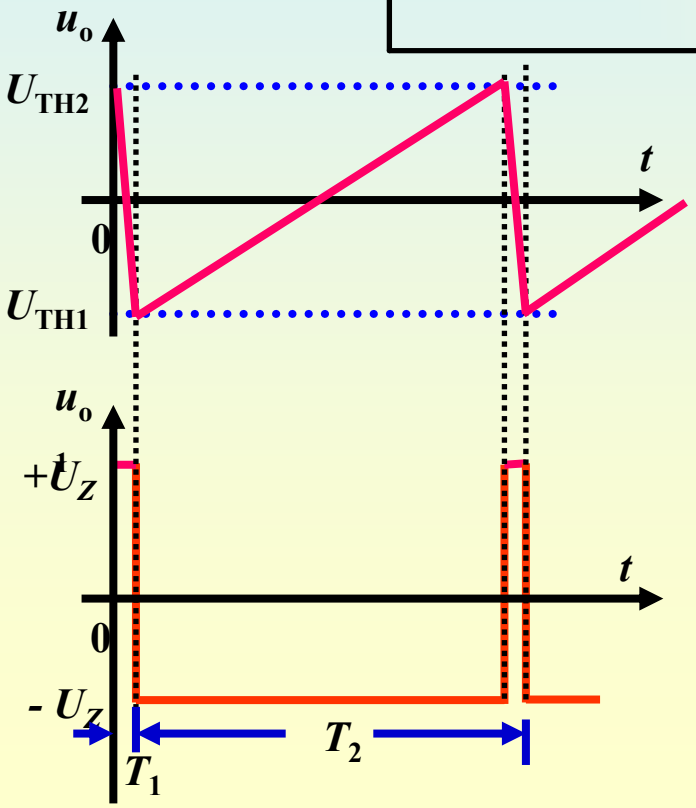




积分时间常数τ不同，u_{o1}矩形波改变了占空比，u_{o2}形成锯齿波。

可调锯齿波发生电路



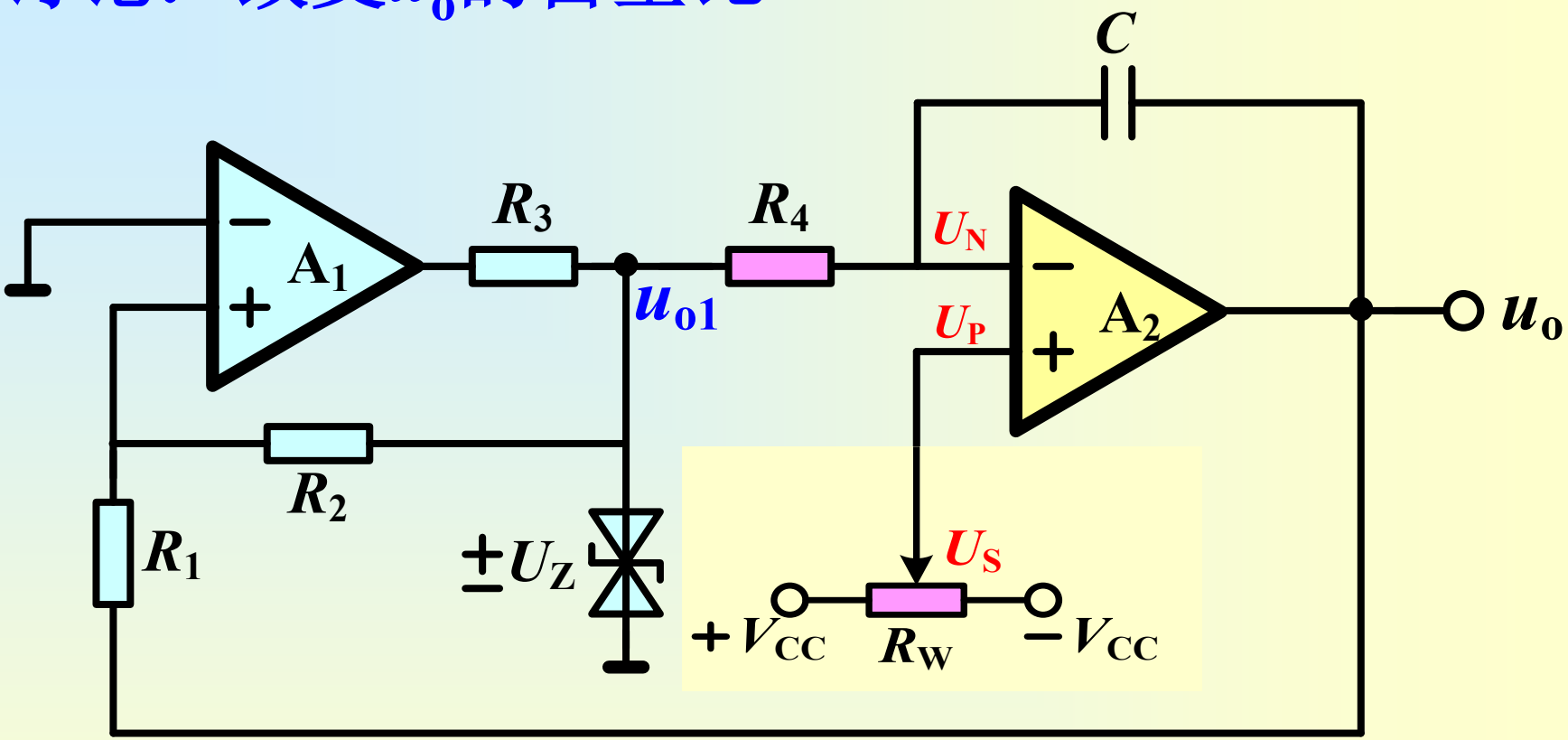


电位器滑动端移到两端时：

$$T_1 \approx 2\frac{R_1}{R_2}R_4C$$

$$T_2 \approx 2\frac{R_1}{R_2}(R_4+R_w)\ C$$

讨论：改变u_o的占空比？



根据“虚短”$U_N=U_S$，使得积分器充电电压(=$+U_Z-U_S$)和放电电压(=$-U_Z-U_S$)不同，也能使u_o形成锯齿波。

非正弦波发生电路小结

1．当运放开环工作或引入正反馈时，运放工作在非线性状态。其分析方法为：

若$U_+ > U_-$　则$U_O = +U_{OM}$；

若$U_+ < U_-$　则$U_O = -U_{OM}$。

虚断（运放输入端电流≈0）

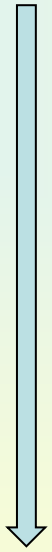
2．比较器是一种能够比较两个模拟量大小的电路。滞回比较器具有回差特性。它们是运放非线性工作状态的典型应用。

3．在矩形波、三角波和锯齿波等非正弦波信号发生器中，核心运放工作在非线性状态。电路由滞回比较器、定时或积分器等环节组成。

非正弦波发生电路练习题

二版教材：第8章 自我检测：8-12；习题：8。

三版教材：第8章 自我检测：9-15；习题：10。



复习好了可以挑战下一页的题目

非正弦波发生电路比较全面的练习题目

选择①增大、②不变或③减小填入空内：

当R_1增大时，u_{o1}的占空比将___，振荡频率将___，u_{o2}的幅值将___；若R_{W1}的滑动端向上移动，则u_{o1}的占空比将___，振荡频率将___，u_{o2}的幅值将___；若R_{W2}的滑动端向上移动，则u_{o1}的占空比将___，振荡频率将___，u_{o2}的幅值将___。

